Option Pricing Using Fractional Order Models and Machine Learning Technique: An Application to the Sri Lankan Financial Market

Bachelor of Science Honours Degree in
Financial Mathematics and Industrial Statistics

By
S.S.Liyanage
SC2020/11797

Supervisor:
Dr. L. W. Somathilake

Department of Mathematics
University of Ruhuna
Matara

Sep 29, 2025

Declaration

I declare that this "Project Report" has not previously been accepted in substance for any degree at the University of Ruhuna or any other institution. Except where otherwise indicated or acknowledged, I declare that this report is the result of my independent work under the supervision of Dr. L. W. Somathilake Where other published and unpublished source materials have been used, these have been acknowledged.

Student Name: Sachini Sandalika Liyanage
(SC/2020/11797)
Department of Mathematics
Faculty of Science
University of Ruhuna
Matara
_____ / _____ / 2025

Supervisor: .....
Department of Mathematics
Faculty of Science
University of Ruhuna
Matara
_____ / _____ / 2025

Acknowledgment

To begin with, I would like to say a very warm thanks to my supervisor, Dr. L. W. Somathilake, who was incredibly helpful, provided constructive feedback and support during the entire course of this research. Their guidance and knowledge have been of critical importance in developing this thesis.

Personally, I would also like to thank the academic faculty of Science at University of Ruhuna as its source of the knowledge base and resources upon which this work is based. I would also like to thank my colleagues and friends who have supported me, discussed some interesting points, and helped me during difficult moments.

I would like to thank my family who have loved me unconditionally, been patient and encouraged me. They have always believed in me and that is what has been pushing me to achieve success in my academics.

Lastly, I consider the Colombo Stock Exchange (CSE) and other publicly accessible financial information sources which offered the critical information in the empirical analysis of the current study. This thesis would not have been completed without these resources.

Abstract

This thesis explores the applicability of the use of fractional Brownian motion (FBM) and its machine learning-enhanced extension (FBM-ML) as alternative option pricing models to the traditional Brownian motion (BM) model in the Sri Lankan equity market. The reason behind this motivation is that BM, the Black-Scholes model, is made under the assumption of independent and normally distributed returns when empirical evidence shows that financial returns are prone to long memory, volatility clustering, and heavy tails. FBM has a generalization of BM, with the Hurst exponent, which permits persistence or anti-persistence in returns, and the machine learning methods provide a method to model time-dependent volatility and memory effects.

The research utilises various volatility estimation; i.e., historical, EWMA, and GARCH-family models, and then the estimation of Hurst exponent through rescaled range analysis, de-trended fluctuation analysis, variance-time scaling, and spectral. BM, FBM and FBM-ML dynamics are estimated using these estimates to simulate the price path. The performance of option prices and hedging is compared with synthetic data (built on the assumptions of BM) and live market data of Indian NIFTY options, which is used as a proxy due to lack of liquid options market in Sri Lanka.

The findings indicate that BM performs better than FBM and FBM-ML in the simulations and this is partly due to the fact that the data generating process is BM based. Nonetheless, FBM is shown to be beneficial when used with real option data, and it is able to model persistence and clustering of volatility, albeit with an overpricing bias at high strikes. The results are in line with previous literature, such as Xu and Yang (2021), which highlight the superiority of FBM in markets that are affected by memory. The thesis concludes by finding that though BM is a viable baseline in Sri Lanka under the current data constraints, fractional models are a superior theoretical and empirical foundation to the option pricing problem in the emerging markets.

Contents

Acknowledgment **ii**

Abstract **iii**

1 Introduction **2**

1.1 Background of the Study 2

1.2 Problem Statement 4

1.3 Aims 4

1.3.1 Objectives 5

1.4 Significance of the Study 5

2 Literature Review **6**

2.1 Classical Option Pricing Models 6

2.2 Fractional Calculus in Finance 7

2.3 Machine Learning in Option Pricing 7

2.4 Hybrid Approaches 8

3 Materials and Methods **10**

3.1 Data 10

3.2 Methods 11

3.2.1 Theoretical Background 11

3.2.2 Data Preprocessing 16
3.2.3 Parameters Estimations 16
3.2.4 Evaluation of Volatility Estimates 19
3.2.5 Model Comparison: Brownian Motion vs. Fractional Brownian Motion 25
3.2.6 Hybrid FBM Model with Machine Learning 29

4 Results and Discussion **34**
4.1 Descriptive Statistical Analysis 34
4.1.1 Volatility Clustering and Time-Series Patterns: 36
4.1.2 Long Memory in Volatility 38
4.1.3 Price Trends and Comparative Behaviour 39
4.2 Volatility and Long-Memory Estimation 40
4.2.1 Volatility Estimation 40
4.2.2 Hurst Estimation 41
4.3 Option Pricing Performance 43
4.3.1 Option Pricing Performance: BM vs. FBM 43
4.4 Hybrid FBM-ML Model Evaluation 44
4.5 Forecast Accuracy Comparison via Diebold–Mariano Test 46
4.6 Model Validation Against Market Data 47

5 Conclusion and Future improvements **49**
5.1 Discussion 49
5.2 Conclusion 50
5.3 Future Improvements 51

Appendix **53**
.1 codes 55

List of Tables

4.1 Best Volatility Estimator per Stock 40

4.2 Simulation Validation of Hurst Estimators 42

4.3 BM vs. FBM Hedging Error Comparison (using best volatility) 43

4.4 Comparison of BM, Classical FBM, and Hybrid ML–FBM Option Pricing Models 45

4.5 Diebold–Mariano Test Results: Model Comparisons 46

4.6 Comparison of Model Prices and Market Prices (NIFTY Options) 48

1 Summary Statistics of Selected Stocks 53

2 Descriptive Statistics and Test Results for Selected Stocks 53

3 Hurst Exponent Estimates for Selected Stocks 53

List of Figures

4.1 Daily closing values of each company 35

4.2 The return distributions 36

4.3 Daily log returns 37

4.4 Volatility Analysis 38

1 Structure of the Data 54

2 normality 54

3 ACF Result 55

Chapter 1

Introduction

1.1 Background of the Study

In the financial market, derivatives are called financial instruments whose value depends on the performance of an underlying asset or index. They serve several purposes, including hedging against risks, speculation, and arbitrage opportunities. Derivatives are mainly of four types: forward contracts, futures contracts, options, and swap contracts. Among these various types of derivatives, options play an important role in financial markets, allowing investors to hedge risk, speculate on price movements, and optimize returns. Options have non-linear payoffs, giving the market participants flexibility in managing exposure to price fluctuations [11]. Options are highlighted contracts that give the holder the right, but not the obligation, to buy (call options) or sell (put options) an asset at a prearranged price before a specified expiration date. Options can be further classified into American options, which can be exercised at any time before expiration, and European options, which can only be exercised at maturity [11].

The derivatives market in Sri Lanka is underdeveloped and the foreign exchange segment is the one that is mainly active. According to the Central Bank of Sri Lanka, businesses hedge against exchange rate fluctuations (such as changing rates between the Sri Lankan Rupee (LKR) and the US Dollar (USD)) using instruments such as forwards, swaps and options [41]. As an example, through currency options, exporters are able to hedge against the appreciation of the rupee and therefore guard their profit margins. Tea is one of the most significant exports that Sri Lanka has in the commodity sector; however, research

indicates that the lack of futures and options contracts restricts the potential of producers to effectively deal with price volatility [46]. The introduction of these tools, such as put options on tea, would give farmers and exporters a tool to hedge against low prices in times of low demand. Though the stock and index options are not actively traded at the Colombo Stock Exchange, the possibility of such markets being developed in the future would be beneficial in managing risks and speculating among the equity investors. All these illustrations demonstrate the present and potential use of derivatives in risk management and strategic decision making in the emerging financial market in Sri Lanka. [51].

Black Scholes, and Merton (1973) introduced the Black Scholes Merton (BSM) model, and this was a breakthrough in option pricing as it offered a closed-form solution to European options [8]. The model is based on the assumption that asset prices move according to Geometric Brownian Motion (GBM), which is constant volatility, log normal returns, and independent price variation over time [11]. This framework enabled investors and institutions to effectively price options methodically and integrate them into risk management plans, which is the basis of contemporary financial theory and practice.

Although it has been widely applied, the BSM model is faced with a number of limitations. Empirical evidence shows that asset returns are frequently skewed, fat-tailed, volatility clustering, and long-term memory; thus, the model fails to satisfy the assumption of the model [34, 13, 33]. Consequently, the BSM model is useful in pricing out-of-the-money options and generates systematic risk assessment biases [22]. Such deviations are especially high in underdeveloped markets, in which volatility is increased by lower liquidity and market inefficiencies, which underscore the pressure to adopt a more flexible pricing strategy.

These limitations have been further emphasized by the current market developments. In April 2020, briefly, crude oil futures went to negative values, which led to a temporary adoption of the Bachelier model in the Chicago Mercantile Exchange [15]. These incidents indicate that pricing structures should be capable of managing extreme movements, nonlinear dynamics and long-range dependence.

To manage these challenges, two approaches have cropped up. The Hurst exponent (H) is used in fractional calculus and fractional Brownian motion (fBM) to model long-term memory in finance time series [35, 52] . Compared to classical models, fractional models such as the fractional Black-Scholes model are more effective in describing volatility clustering and heavy tails [50, 12]. Machine learning (ML) is a technology capable of

modelling nonlinear relationships with complex characteristics in option pricing without any limiting assumptions [25, 54, 49, 15]. The properties available in ML models, such as realized volatility, regime switches, and fractal diagnostics, provide flexible and adaptive prediction abilities.

These issues are further exacerbated in Sri Lanka because there is a lack of liquidity and derivative products on the Colombo Stock Exchange (CSE). The construction of synthetic options based on realized volatility or proxy options in close markets, such as India, is a convenient method for simulating and testing advanced pricing models [43, 53].
Existing research concentrates on developed markets, and there is an evident gap that needs to be filled in research on emerging economies. This hybrid method aims at giving a more precise pricing and better risk management strategies to the local funders and institutions by modelling long-memory effects, anomalous diffusion, and volatility clustering.

1.2 Problem Statement

Why are the classical models, such as the Black-Scholes Model, broken down in certain financial market conditions (ex., when there is long memory or persistence in asset prices: here consider stock option prices as an asset price)

1.3 Aims

This research study aims to develop an option pricing model using Fractional calculus methods and Machine Learning techniques for synthetic options based on historical stock data from the Colombo Stock Exchange (CSE). This will help address the pricing challenges posed by Sri Lanka's developing market conditions [15, 18].

1.3.1 Objectives

- Investigate the more accurate options pricing model using suitable fractional calculus methods and machine learning techniques.
- Focus on the use of fractional Brownian motion (FBM) for stock option price dynamics and machine learning techniques for parameter estimation(eg: Hurst, volatility).
- Develop and apply these models in the context of the Sri Lankan financial market.

1.4 Significance of the Study

This study modified traditional option pricing models by integrating fractional calculus and machine learning to account for current financial market behaviours like volatility swing and market memory. The findings provide better pricing and risk management strategies, particularly for the Sri Lankan financial market.

The remainder of this study is organized as follows: Chapter 2 reviews existing literature on their applications :reviews traditional option pricing models, focusing on the Black Scholes framework and its limitations. Chapter 3 describes the data sources, synthetic option construction, and methodological framework. Chapter 4 discusses the results and comparative analysis. Chapter 5 concludes with implications, limitations, and future research directions.

Chapter 2

Literature Review

2.1 Classical Option Pricing Models

The modern option pricing theory originated with the black-scholes model (BSM) and its variation by Merton (1973), which was a closed-form solution of European options. The BSM assumes that the stock prices are governed by a geometric Brownian motion that has a constant volatility and risk-free rates in a frictionless market [8]. This innovation defined a vast part of the financial economics field and is still widely applied as a benchmark.

Even with its impact, the BSM has faced limitations. It has been observed using empirical data that real markets exhibit volatile smiles, heavy tails, and sudden leaps, which none of the assumptions of the model can explain [13, 34]. These inadequacies led to more long-term models such as the Heston stochastic volatility model, in which volatility may change over time, and jump-diffusion models such as Bates (1996), in which volatility may jump [22, 6]. Other models, such as the Levy processes and models with infinite variance, are also trying to explain the irregular returns [52]. However, these models may be sophisticated to calculate and less feasible in an emerging market with limited data [43].
Classical models' weaknesses are more evident when contrasted with stylized facts of financial data. Studies have indicated that returns normally exhibit:

- Volatility clustering - turbulent periods are likely to occur in series [13].
- Fat tails and skewness - large movements of prices are more frequent than they

should be according to the normal distribution [34].

- Long memory - volatility is persistent across long periods of time [12, 33].
- Effect of leverage - the volatility increases more when the price falls [13].

2.2 Fractional Calculus in Finance

To overcome these limitations, Mandelbrot and Van Ness (1968) proposed the idea of fractional Brownian motion (fBM), where the long-range dependence is added to the concept of Brownian motion [35]. This was supported in later studies, which reported the view that fractional processes are quite appropriate in explaining market persistence and irregular price movements [50, 11]. They are particularly significant in frontier and emerging markets where prices can be described as skinny, structural, and external shocks [53, 46, 51].
Fractional calculus is a generalization of derivatives to non-integer orders that enable models to capture memory effects and anomalous diffusion. Fractional derivatives differ from standard calculus, which takes into account only local changes, unlike the use of the past states in current dynamics. This is especially applicable in modeling persistent volatility and long-memory in returns in finance.

Fractional processes were demonstrated to enhance stochastic volatility models by Comte and Renault (1998) [12]. In Hu and Ouerdane (2011), the authors showed the long-range dependence of financial series using the fractional Brownian motion [11]. Most recently, Raubitzek, Mallinger, and Neubauer (2023) surveyed the application of machine learning with fractional calculus, finding such models to be more effective in irregular volatility, and particularly useful in volatile markets [55].

In the case of such a country as Sri Lanka, where the volatility is substantial and the financial data do not necessarily conform to the conventional assumptions, option pricing can be built on a more solid basis with the help of the fractional methods [41].

2.3 Machine Learning in Option Pricing

Simultaneously, machine learning (ML) has become a significant option pricing instrument. Contrary to the parametric models, ML is data-oriented and it is able to learn

non-linear patterns directly with large datasets.
One of the examples, Hutchinson, Lo, and Poggio (1994) [25], used neural networks to price options and demonstrated better performance as compared to Black-Scholes. A variety of methods have been applied since then:

- Support Vector Machines (SVMs) - useful in non-linear regression and in high-dimensional problems [56, 26].
- Ensemble models like Gradient Boosting and random forests - handy when you have structured option data [49].
- Deep learning approaches such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks - created to encode sequential and time-dependent structures [54].

Empirical research always indicates that ML models are more effective than BSM. An example is where Gross in 2022 compared Black-Scholes, Bachelier, and neural networks and found that it was definitely improved by ML [15]. Chowdhury et al. in 2020 used modified Black-Scholes models with ML in the frontier markets and got a better predictive power [47]. Likewise, Ruf and Wang conducted a review of neural networks in pricing and hedging of options and found that it has benefits in terms of their accuracy and capability [49].

2.4 Hybrid Approaches

Recent research has shifted towards hybrid models combining theoretical and data-driven approaches. The reasoning is that fractional models represent the long-term memory and structural dependencies, whereas machine learning can adapt to the short-term and non-linear changes. Raubitzek et al. (2023) identified hybrid fractional-ML models as one of the promising future directions in financial studies [55]. In the case of emerging markets, as depicted by Singh and Patel, the conventional models have been found to fail in most cases, and hybrid models are therefore more feasible [53]. Sri Lanka's commodity markets, such as tea futures, have case studies that reveal the significance of modelling, which considers local dynamics [46, 51].
Regulators and policymakers can also make use of hybrid approaches. According to reports of the Colombo Stock Exchange and the Central Bank of Sri Lanka (2023), there

is a potential threat of exchange rate shock, liquidity constraints, and limited derivatives [18, 41]. Combination models of fractional derivatives and ML are a flexible and powerful tool to study these conditions.
Fractional calculus and ML are historically researched in developed markets, but frontier markets are barely studied [54, 49]. Particularly, India has been a focus of some attention, but Sri Lanka is virtually unknown [53]. This is because the absence of an active options market necessitates scholars to use synthetic option construction.

So, with the limitations of the classical models and the strengths of FBM and machine learning, this research hopes for a hybrid approach that combines fractional calculus with machine learning to model stock options in the Sri Lankan financial market. This hybrid model aims to improve pricing accuracy by accounting for long-term memory dependencies and market anomalies, leveraging machine learning for more precise parameter estimation.

Chapter 3

Materials and Methods

3.1 Data

In this study, the empirical analysis relies on past stock price records of the few companies that are listed on the Colombo Stock Exchange (CSE). The sample covers companies in the different sectors of the industry, like banking and finance, telecommunications, manufacturing, healthcare, and diversified holdings. The companies were selected due to their high market capitalization, reasonably high trading activity and representing their respective sectors; thus, they can be considered a perfect proxy of the Sri Lankan equity market.
Data for the period November 2014 to November 2024 were gathered, in terms of daily close prices, trading volume, and other market data makers, on the official CSE database and other reputable sources of financial data.
As there is no active options market in Sri Lanka at present, synthetic options data were built on the basis of underlying stock price data. The data of the Indian derivatives market (NIFTY) options market was used as a benchmark to make it robust and comparable, especially the actively traded equity options. The data has strike price, maturity, implied volatility and interest, allowing the calculation of benchmark Black -Scholes option prices and calibration to models. [18].

This data set was composed of two major elements:

- This Sri Lankan equity data:- daily returns and prices of five companies of choice, applied to long-term memory estimating and fractional calculus modeling.

- Synthetic option data:– This was constructed based on Sri Lankan stock prices and benchmarked against Indian equity options to evaluate the performance of the introduced pricing models in an emerging market context.

3.2 Methods

This research will focus on developing an option pricing model using a combination of Fractional Calculus methods with Machine Learning (ML)techniques to improve the accuracy and strength of stock option prices in developing market such as Sri Lankan financial market.

3.2.1 Theoretical Background

Brownian Motion Models and Assumptions

Classical financial models assume asset prices follow a Brownian motion process with specific idealized conditions. In the Black–Scholes framework, prices are modeled by a geometric Brownian motion (GBM) with continuous trading and no arbitrage opportunities. This implies independent, memoryless price changes; the process has the Markov property, meaning future price movements depend only on the current price and not on the path taken (the process has 'no memory') [45]. Equivalently, price increments are assumed independent and stationary, so that past fluctuations do not influence future volatility or drift [45]. Volatility is treated as a constant parameter, known in advance, and the risk-free interest rate is also constant [21]. These assumptions underlie the efficient market hypothesis, where in-price movements are essentially unpredictable random walks [21, 23]. It means that the "Black-Scholes world" assumes a frictionless market with no transaction costs, continuous trading, and log-normal price dynamics with constant drift and volatility [21]. Under these conditions, asset returns are normally distributed and uncorrelated over time, consistent with an information-efficient market [21]. Such a model (with Hurst exponent $H = 0.5$) exhibits no long-term memory by construction [2].

Real-world data, especially from emerging or developing markets often violate the above assumptions. Empirical studies document market inefficiencies and predictable patterns in developing markets that contradict the pure random walk hypothesis. For instance, variance ratio tests have rejected random walk behavior in several Asian emerging mar-

kets, indicating that returns can exhibit short-term auto-correlations (momentum or mean-reversion), inconsistent with strict efficiency [24].

A well-known stylized fact is volatility clustering, first noted by Mandelbrot (1963): large price changes tend to be followed by large changes (of either sign), and small changes by small changes .In quantitative terms, while raw returns in liquid markets may show little auto-correlation, the absolute or squared returns display positive, slowly decaying auto-correlation, implying persistence in volatility. This long-lasting dependence on volatility contradicts the constant volatility assumption of GBM. It has been observed across markets that volatility is heteroskedastic and mean-reverting rather than constant [59].

Emerging markets often exhibit even stronger volatility persistence and long memory effects than developed markets [7, 24]. Long memory refers to significant correlations at long lags; for example, the Hurst Mandelbrot rescaled range analyses of emerging market indexes reveal $H > 0.5$ in many cases, indicating persistent dynamics [24]. These markets may also suffer from lower liquidity and information inefficiency, which contribute to slowly decaying auto-correlations and predictable volatility patterns(violating the 'memory-less' assumption). Indeed, long-range dependence in volatility is now recognized as a ubiquitous phenomenon–financial time series across many markets show that volatility has a hyperbolic decay of auto-correlation, inconsistent with the exponential decay assumed by simple GARCH models [58]. Such evidence has motivated extensions of the GBM model to include stochastic volatility and other features. In particular, ARCH/GARCH models were introduced to capture volatility clustering and excess kurtosis [9, 16]. These models allow volatility to depend on past shocks, rather than remain constant, there by reproducing the clustering effect. Likewise, observed leverage effects (the tendency for negative returns to increase future volatility more than positive returns) violate the symmetry assumed in basic GBM. This asymmetry is especially relevant in emerging markets during downturns. To address it, asymmetric GARCH variants such as EGARCH (Nelson, 1991) and GJR-GARCH (Glosten et al., 1993) were developed [20, 39]. EGARCH uses a logarithmic volatility equation to ensure positivity and incorporates a term for the sign of returns, allowing negative shocks to have a larger impact on volatility [40].

Similarly, the GJR-GARCH model includes an indicator for negative returns so that 'bad news' increases conditional volatility more than 'good news' of equal magnitude, captur-

ing the leverage effect observed empirically [57].All these empirical deviations – volatility clustering, long memory, and leverage effects are pronounced in frontier and developing markets like the Colombo Stock Exchange, undercutting the standard Black–Scholes assumptions of independent increments, no memory, and constant volatility.

Fractional Brownian Motion (FBM)

Fractional Brownian motion, introduced by Mandelbrot and Van Ness (1968), generalizes classical Brownian motion by incorporating memory through the Hurst exponent H [35]. FBM $B_H(t)$ is a zero-mean Gaussian process with continuous paths and covariance [2, 52].

$$E[B_H(t)B_H(s)] = \frac{1}{2}(t^{2H} + s^{2H} - |t-s|^{2H}), \qquad s, t \geq 0 \tag{3.1}$$

One useful definition of FBM is in terms of the Mandelbrot-Van Ness moving-average representation:

$$B_H(t) = \int_{\mathbb{R}} \left[(t-u)_+^{H-\frac{1}{2}} - (-u)_+^{H-\frac{1}{2}} \right] dW_u, \tag{3.2}$$

Where, dW_u, is a two-sided Brownian motion. The covariance of (3.1) is reproduced by this type of kernel form [35, 52].

Unlike standard Brownian motion (which has $H = 0.5$), an FBM with $H \neq 0.5$ exhibits dependent increments. Specifically, if $H > 0.5$, the increments have positive long-range correlation (persistence): a high return is likely to be followed by another high return (of either sign), generating long memory in the series [2]. If $H < 0.5$, increments are negatively correlated (anti-persistent), producing a mean-reverting or short-memory effect . In the special case $H = 0.5$, FBM reduces to ordinary Brownian motion with independent increments (no memory) [2] .
Fractional Gaussian noise (FGN) is called as the increment process here:

$$X_t = B_H(t+1) - B_H(t), \tag{3.3}$$

where $B_H(t)$ is fractional Brownian motion.

The autocorrelation function behaves as

$$\rho(k) \sim H(2H-1)\, k^{2H-2}, \qquad k \to \infty, \tag{3.4}$$

showing long memory when $H > 0.5$ and anti-persistence when $H < 0.5$ [13, 7].
Fractional Gaussian noise (the increment process of FBM) can thus model the empirically observed long-range dependence in financial data. The self-similarity property of FBM (scaling behaviour of increments) aligns with the fractal characteristics of market time series, as described in the Fractal Market Hypothesis. By introducing the Hurst exponent, FBM provides an additional degree of freedom to capture the memory and correlation structure of asset returns [2] [33, 58].

FBM is self-similar it's mean:

$$B_H(at) \overset{d}{=} a^H B_H(t), \qquad a > 0, \tag{3.5}$$

which implies that [35, 52] ,

$$\mathrm{Var}\Big[B_H(t)\Big] = t^{2H}. \tag{3.6}$$

In developing markets where H is often significantly different from 0.5, an FBM-based model can reflect the persistent volatility or trending behavior better than a memoryless GBM. In terms of modeling asset prices, replacing the standard Brownian motion $W(t)$ with a fractional Brownian motion $B_H(t)$ in the SDE yields a geometric fractional Brownian motion (GFBM) model. The GFBM dynamics can be written as:

$$dS_t = \mu S_t\, dt + \sigma S_t\, dB_H(t) \tag{3.7}$$

The harmonizable representation is also available in FBM in the frequency domain:

$$B_H(t) = C_H \int_{\mathbb{R}} \frac{e^{it\xi} - 1}{|\xi|^{H+\frac{1}{2}}}\, \widehat{W}(d\xi), \tag{3.8}$$

where $\widehat{W}(d\xi)$ is a complex Gaussian white-noise measure. This is particularly convenient in FFT-based simulation (e.g., DaviesHarte) [52, 12].
with solution $S_t = S_0 \exp\Big(\mu t - \frac{1}{2}\sigma^2 t^{2H} + \sigma B_H(t)\Big)$ [2]. Here H, governs the roughness of the path and the growth rate of variance (for $H = 0.5$, $t^{2H} = t$ recovers the GBM solution). The advantages of this formulation lie in its ability to reproduce stylized facts: for $H > 0.5$, simulated returns will exhibit long-term correlations and a variance that grows faster than linearly with time, echoing the volatility clustering/long memory seen in practice [13, 7].
Empirical research has indeed found that incorporating fractional dynamics can improve

the realism of price simulations and forecasts. For example, numerous studies have detected memory in asset returns and volatilities that GBM cannot account for by using FBM, the model 'remembers' past volatility, producing the slowly decaying auto-correlation in absolute returns observed in emerging markets [32, 58]. In forecasting tasks, including a Hurst term has yielded better out-of-sample performance since FBM can adapt to persistent trends or cycles in the data [2].

Fractional models address volatility dynamics by naturally generating a variance time scaling of

$$Var[\Delta S] \propto \Delta t^{2H}, \tag{3.9}$$

as opposed to the linear scaling ($H = 0.5$) in standard diffusion [4, 12]. This helps capture the empirically observed deviation from the square-root of time rule for volatility accumulation in markets with long memory. It is important to note that FBM is not a semi-martingale for $H \neq 0.5$, which historically raised questions about arbitrage in fractional models [38] . Early critiques (e.g. Rogers, 1997) showed that a frictionless market driven by FBM would allow arbitrage opportunities due to the long-range dependence. However, subsequent work introduced alternative stochastic integration frameworks to construct arbitrage-free fractional pricing models [38, 48]. Hu and Oksendal (2000) demonstrated that if one assumes realistic market frictions or discrete trading (finite reaction speed of investors), arbitrage can be avoided and a fractional Black–Scholes formula can be derived . In this thesis, we focus on the statistical modeling advantages of FBM (capturing memory and volatility clustering) rather than its arbitrage implications. The FBM-based approach is particularly appealing for developing markets like Sri Lanka's, where strong persistence and deviations from classical assumptions are evident. By combining FBM with modern estimation techniques, we aim to better characterize the price dynamics and improve option pricing and hedging performance in such markets

3.2.2 Data Preprocessing

Return Computation

In line with standard practice in financial econometrics, we convert the price series into logarithmic returns to stabilize variance and obtain approximately stationary time series [58]. If P_t is the adjusted closing price on day t, the log return is computed as:

$$r_t = \ln(P_t/P_{t-1}) = \ln P_t - \ln P_{t-1}. \tag{3.10}$$

These log returns r_t serve as the primary input for volatility and Hurst exponent estimations. Log returns are advantageous because their distribution is closer to normal (compared to raw price changes) and the compounding effect is handled additively.

Stationarity and Distribution Checks

After computing log returns, we conducted preliminary statistical tests to verify the series' suitability for modeling. An Augmented Dickey–Fuller (ADF) test was applied to each stock's return series, and in all cases the null hypothesis of a unit root was strongly rejected, confirming that the return series are stationary (zero mean after removing any small drift) [58].The return distributions exhibited heavy tails and skewness (e.g. average excess kurtosis ≈ 4.0, skewness $\approx$–0.5), deviating from normality [58]. These distributional characteristics will be accounted for in model evaluation (e.g. via the Kolmogorov–Smirnov test in the model comparison section).

3.2.3 Parameters Estimations

Volatility Estimation

We implement multiple volatility estimation techniques, ranging from simple historical measures to sophisticated econometric models. Volatility here refers to the conditional standard deviation of returns over a given horizon (daily in our case). The following methods are applied to each stock's return series:

- Rolling Window Standard Deviation: A simple historical volatility estimator, computed as the standard deviation of returns over a recent fixed window (e.g. 30 trading days). Specifically, for each day t (after the first 30 days), we calculate

$$\hat{\sigma}t^{(\text{roll})} = \sqrt{\frac{1}{n-1}\sum^{t}(r_i - \bar{r}t)^2} \tag{3.11}$$

using a window of $n = 30$ days, where $\bar{r}$ is the mean return in that window (often close to 0). This rolling volatility captures local fluctuations but can be noisy. We chose a 30-day window to reflect approximately 1.5 months of trading, a common choice in practice for short-term volatility [11]. The rolling estimate updates slowly as the window moves, dropping the oldest observation each day and adding the newest. While easy to implement, this method assumes volatility is piecewise constant within each window and responds with a lag to sudden shocks.

- Exponentially Weighted Moving Average (EWMA): To give more weight to recent observations, we use EWMA volatility as popularized by J.P. Morgan's RiskMetrics (1996) [29]. The EWMA forecast for variance on day t is

$$\hat{\sigma}t^{2(\text{EWMA})} = \lambda\,\hat{\sigma}^{2(\text{EWMA})} + (1-\lambda)\,r_{t-1}^2 \tag{3.12}$$

where λ is a smoothing parameter. Risk Metrics recommends $\lambda = 0.94$ for daily data [1, 13], implying roughly 94% of the previous day's variance carries forward and 6% of new information comes from yesterday's return shock. We adopt $\lambda = 0.94$in our implementation. This yields a volatility estimate.

$$\sigma t^{(\hat{\text{EWMA}})} = \sqrt{\hat{\sigma}}^{2(\text{EWMA})} \tag{3.13}$$

The EWMA captures volatility clustering by assigning persistence to volatility: a large return (high r^2) will boost the variance estimate, which then decays exponentially over subsequent days if no further large shocks occur. Unlike the rolling window, EWMA effectively has an infinite memory (with geometrically decaying weights) and adapts faster to regime changes. It assumes volatility mean-reverts to some long-run level implicitly given by the average of past squares.

- GARCH(1,1) Model: We fit the Generalized Auto-regressive Conditional Heteroskedasticity model of Bollerslev (1986) to each return series [16].

The GARCH(1,1) specification is:

$$r_t = \mu + \varepsilon_t, \tag{3.14}$$

$$\varepsilon_t = \sigma_t z_t, \tag{3.15}$$

with $z_t \sim N(0,1)$ i.i.d., and

$$\sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2 \tag{3.16}$$

where $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$ and $\alpha + \beta < 1$ to ensure stationarity. This model posits that today's variance σ_t^2 depends on a constant term, yesterday's squared shock (ε_{t-1}^2), and yesterday's variance. It can capture volatility clustering by creating a feedback loop: a big shock raises ε_{t-1}^2, which increases σ_t^2, meaning volatility will be higher today; if no further shocks occur, σ^2 gradually decays towards $\omega/(1-\alpha-\beta)$ over time. We estimate GARCH(1,1) parameters via maximum likelihood for each stock's returns. Typically, we expect α (ARCH term) to be significant (news impact) and β to be large (volatility persistence), reflecting that these markets have high persistence in volatility. GARCH is a benchmark model for conditional heteroskedasticity and often provides a good first-order fit to financial volatility [10].

- EGARCH(1,1) Model: We also implement the Exponential GARCH model (Nelson, 1991) to account for asymmetry in volatility response [39]. EGARCH specifies the log-variance as:

$$\ln \sigma_t^2 = \omega + \phi\, \ln \sigma_{t-1}^2 + \alpha \left(\frac{|\varepsilon_{t-1}|}{\sigma_{t-1}} - \frac{\sqrt{2}}{\pi} \right) + \gamma\, \frac{\varepsilon_{t-1}}{\sigma_{t-1}}, \tag{3.17}$$

  where γ is the leverage coefficient. In this model, volatility responds to both the magnitude and sign of the last shock. A negative ε_{t-1} can increase volatility more if $\gamma < 0$ (typically the case for equities, indicating leverage effect) . The use of $\ln \sigma^2$ guarantees σ_t^2 stays positive without needing parameter constraints. We estimate EGARCH(1,1) for each series (using a Student-t error distribution if necessary to account for fat tails) [40]. The EGARCH model allows us to see if accounting for asymmetry significantly improves volatility forecasts for CSE stocks for instance, if bad news tends to spike volatility more than good news, we expect γ to be significantly negative.

- GJR-GARCH(1,1) Model: Lastly, we fit the Glosten–Jagannathan–Runkle GARCH

model (1993), another asymmetric variant [20]. Its variance equation is:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \gamma\varepsilon_{t-1}^2\mathbf{1}_{\{\varepsilon_{t-1}<0\}} + \beta\sigma_{t-1}^2 \tag{3.18}$$

Here, $\mathbf{1}_{\{\varepsilon_{t-1}<0\}}$ is an indicator function that equals 1 if the previous return shock was negative. If $\gamma > 0$, a negative return increases next-day volatility by $\alpha+\gamma$ (as opposed to α for a positive return) [57]. This model directly parametrizes the leverage effect: one expects $\gamma > 0$ in equity markets.
We estimate $(\omega, \alpha, \gamma, \beta)$ for each stock via maximum likelihood. The GJR-GARCH provides a more flexible form than EGARCH in that it keeps the variance in levels (not log) but penalizes negative shocks. It is useful to compare which asymmetric model (EGARCH or GJR) fits better, as this can indicate the nature of asymmetry in the Sri Lankan context.

3.2.4 Evaluation of Volatility Estimates

We compare the accuracy of these volatility estimators using standard error metrics. The realized volatility on day t is proxied by the observed absolute return or squared return $|r_t|$ (or r_t^2) since we lack intra-day data for a precise realized volatility, the squared daily return serves as a noisy indicator of the true volatility [17]. Using the test sample, we compute the root mean squared error (RMSE) and mean absolute error (MAE) between the predicted volatility (or variance) and the realized volatility (or variance). For instance, if $\hat{\sigma}t$ is the forecast and the realized volatility is $|r_t|$, we compute

$$RMSE = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(\hat{\sigma}_t - |r_t|)^2} \tag{3.19}$$

$$MAE = \frac{1}{T}\sum_{t=1}^{T}|\hat{\sigma}_t - |r_t|| \tag{3.20}$$

(for variance forecasts, r_t^2 would be used). These metrics gauge average prediction error; lower values indicate better forecasting performance. We also consider the $|\sigma\hat{bias}|$(mean error) to see if a model consistently over or under predicts volatility. Given prior evidence in other markets, we expect that GARCH-type models will outperform the naive rolling and EWMA in RMSE/MAE , due to their ability to mean-revert and adapt to shock impacts [14]. The asymmetric models (EGARCH, GJR) may further improve accuracy if

leverage effects are present in the data. Overall, this multi-model approach provides a comprehensive volatility analysis, and the best-performing volatility model (in terms of forecast error) will inform our subsequent option pricing evaluation (since option Greeks like delta and vega depend on volatility dynamics)

Hurst Exponent Estimation

To quantify the degree of long-range dependence in each stock's return series, we estimate the Hurst exponent (H) using four different methods. The Hurst exponent is a key parameter in fractional Brownian motion, and it indicates the presence of memory: $H = 0.5$ corresponds to no long-term memory (random walk), $H > 0.5$ indicates positive long-range correlation (persistent, trending behaviour), and $H < 0.5$ indicates negative correlation (mean-reversion/anti-persistence) [3, 34, 32].
Estimating H will help us determine whether a fractional Brownian model is warranted for the data and, if so, what value of H to use. The methods employed are:

- Rescaled Range (R/S) Analysis: Originally introduced by Hurst (1951) and later popularized by Mandelbrot, the R/S method computes the rescaled range $R(n)/S(n)$ for the time series over different aggregation lengths n [34]. For a given n, one divides the return series into segments, computes the range (max cumulative deviation minus min cumulative deviation) and divides by the standard deviation of returns in that segment. The average R/S is plotted against n on log–log scale. The Hurst exponent is estimated by the slope in the relation

$$E[R(n)/S(n)] \propto n^H \tag{3.21}$$

  . In practice, we compute R/S for window sizes spanning from about 10 days to 1/4 of the sample length (to have multiple segments), and use a linear regression on $\log n$ vs $\log(R/S)$ to estimate H. We also apply Lo's (1991) bias-correction for short-term dependence by choosing an appropriate bandwidth to remove AR(1) effects, ensuring the R/S statistic is focused on long-term range. R/S analysis is one of the most widely used methods for Hurst estimation in finance [30] . However, it can overestimate H in finite samples and in presence of short-range correlations [30] . We will compare its results with other methods for robustness.

- Detrended Fluctuation Analysis (DFA): DFA is a more modern technique that is robust to trends and non-stationarities in the mean [30].

  1. Form the cumulative sum (profile) of the return series

$$X(t) = \sum_{i=1}^{t} \left(r_i - \bar{r}\right).$$

2. Divide $X(t)$ into segments of length n.
3. In each segment, fit a linear trend (or a polynomial of order m) to $X(t)$ and subtract it.
4. Compute the root-mean-square fluctuation

$$F(n) = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\Big(X(t) - \text{trend}_n(t)\Big)^2}.$$

5. Repeat the procedure for various segment lengths n. The Hurst exponent H is estimated from the power-law relation

$$F(n) \sim n^H$$

using a log–log plot [27, 31].

. We use DFA with linear detrending (DFA-1) for simplicity. DFA is advantageous in that it can distinguish long-term dependence from simple trends: any polynomial trend is removed, isolating the fluctuation scaling. It has been widely applied in financial time series to estimate H and often is considered more reliable than R/S for finite samples, as R/S may be biased upward [30]. We implement DFA on each stock's returns for scales from 10 days to 1,000 days. The slope H is obtained by OLS regression on the log–log plot, excluding the smallest scales where discrete effects dominate [7, 30, 27].

- **Variance-Time Plot (Aggregation Method)**

This method exploits the relation

$$\text{Var}(R_{\Delta t}) \propto (\Delta t)^{2H}$$

for fractional processes. We compute aggregated returns over longer intervals (e.g., 5-day, 10-day, 20-day returns) and calculate their variance. Plotting the sample variance of m-day returns versus m on a log–log scale should yield a line with slope $2H$ if long-range dependence holds [13, 24].

In practice, for each stock we form non-overlapping m-day returns:

$$R_k^{(m)} = \sum_{i=0}^{m-1} r_{k \cdot m+i}, \quad m = 1, 2, 4, 8, \ldots, 128$$

(up to quarters or so given our sample). We then estimate the Hurst exponent H by regressing

$$\log\left(\widehat{\mathrm{Var}}[R^{(m)}]\right) \text{ on } \log m.$$

This is essentially a time-domain rescaling approach and is conceptually simple. If the returns were a short-memory process (e.g., white noise or ARMA), we expect a slope of approximately 1 (since variance scales linearly with time for $H = 0.5$). A slope significantly greater than 1 indicates $H > 0.5$ (persistent volatility), while a slope less than 1 indicates anti-persistence. This method can be sensitive to outliers in variance estimation, so we also examine the variance of absolute returns or use the median absolute deviation for robustness [13].

- **Periodogram (Frequency-Domain) Method**

Long-memory processes have a characteristic spectral density

$$f(\omega) \sim C\,\omega^{1-2H} \quad \text{as } \omega \to 0, \quad 0.5 < H < 1.$$

We apply the periodogram estimator using the GPH method (Geweke & Porter-Hudak, 1983):

1. Take the Fourier frequencies

$$\omega_j = \frac{2\pi j}{T}, \quad j = 1, \ldots, T/2.$$

2. Compute the periodogram $I(\omega_j)$.
3. Regress

$$\log I(\omega_j) \text{ on } \log\left(4\sin^2(\omega_j/2)\right)$$

for the lowest, say, 10% of frequencies.

The slope of this regression is an estimate of the fractional differencing parameter

$$d = H - 0.5,$$

from which the Hurst exponent is obtained as

$$H = 0.5 + \hat{d}.$$

The periodogram method provides a spectral-domain perspective complementary to time-domain methods. It is asymptotically unbiased but can suffer from high variance in finite samples. By comparing periodogram results with DFA and R/S methods, we can increase confidence in our H estimates [4, 12].

After applying all four methods to each stock's return series, we obtain a set of Hurst exponent estimates.We expect emerging market stocks to exhibit $H > 0.5$ in volatility (due to volatility clustering), and possibly$H$ slightly above 0.5 in raw returns (indicating mild predictability or long memory in returns) although many studies find that raw returns (excluding volatility effects) are nearly uncorrelated ($H \approx 0.5$) when adjusting for trends [58]. Indeed, prior research on long memory often finds strong long-range dependence in volatility but not necessarily in returns themselves (the "long memory of volatility" phenomenon) [59]. Validation via Simulation: To ensure the reliability of our Hurst estimation, we perform a validation exercise. We generate synthetic fractional Brownian motion series with known Hurst exponents (e.g. $H = 0.6, 0.75$) using the Cholesky method or Davies–Harte algorithm for length $T \approx 2500$ (matching our data length). Then we apply the above estimation methods to these simulated series to see if they recover the true H. This allows us to gauge the bias and variance of each estimator in a controlled setting. As expected from literature, we found that DFA tends to produce estimates closer to the true H with smaller bias for moderate sample sizes, whereas R/S often yielded slightly higher H than true (overestimation) [30] . The periodogram method had larger variance but was unbiased on average. Based on these results, we place greater confidence in the DFA estimates. DFA's robustness to trends and moderate sample performance makes it a preferred choice in recent literature [30] . Consequently, for each stock we will use the Hurst exponent estimated via DFA as the primary value of H for modelling purposes. These are broadly consistent, supporting the presence of long-term dependence in the data. Thus, we conclude that a fractional Brownian component (with $H \approx 0.6$) is appropriate to include in our option pricing model for these stocks.

3.2.5 Model Comparison: Brownian Motion vs. Fractional Brownian Motion

With the volatility and Hurst parameters estimated, we proceed to compare the dynamics of a standard Brownian motion model (GBM) and a fractional Brownian motion model (FGBM) against the observed data. The goal is to assess which stochastic process better reproduces the statistical properties of the stock prices in our sample. This involves simulation and goodness-of-fit measures focusing on distribution and dependence characteristics: Model Specifications: For each stock, we calibrate two simulation models for the underlying asset price S_t:

- Geometric Brownian Motion (GBM): This is the classical model used in Black–Scholes [8, 36, 11]. We use the estimated daily drift $\hat{\mu}$ (mean return) and volatility $\hat{\sigma}$ (which could be the sample standard deviation or an estimate from the volatility models) to define $dS_t = \hat{\mu} S_t \, dt + \hat{\sigma} S_t \, dW_t$. Because drift is hard to estimate reliably and is very small for daily data, we set $\hat{\mu}$ to the sample average return (which for many stocks is near zero or a few percent annualized) for simulation. The key parameter is $\hat{\sigma}$; we use an appropriate value reflecting average volatility (e.g. we might take the in-sample average of the GARCH-implied volatility or simply the sample standard deviation of returns). This GBM assumes $H = 0.5$ (no memory) and serves as the null model [45].

- Geometric Fractional Brownian Motion (FGBM): Using the DFA-estimated Hurst exponent $\hat{H}$ for the stock and the same volatility $\hat{\sigma}$ (interpreted now as scale parameter in the FBM's increments), we define $dS_t = \hat{\mu} S_t \, dt + \hat{\sigma} S_t \, dB_{\hat{H}}(t)$. We simulate the fractional Brownian motion $B_{\hat{H}}(t)$ with the given H using a discretization (e.g. Cholesky or circulant embedding to generate correlated Gaussian increments). The drift $\hat{\mu}$ is kept the same as in GBM for fairness (though in fractional models drift has less impact on short-term dynamics). Essentially, the FGBM is an "evolution" of GBM that incorporates memory via the parameter H [2, 61, 52, 35].

For each model, we generate a large number of price paths (e.g. 5,000 simulations) over the 10-year horizon to match the sample length. We start each simulation at the actual initial price S_0 of the stock in 2014 and simulate daily steps. For FGBM, we use a discrete approximation that yields the correct covariance structure for increments [48, 38]. Comparison Metrics: We evaluate the simulated price paths against the real price series using several metrics:

- Autocorrelation Function (ACF) of Returns: We compute the sample autocorrelation function ρ_k for log-returns at various lags k (e.g. $k = 1$ up to 30 days) for both the real data and the simulations. For each model, we average the autocorrelations across many simulated paths to get an expected ACF. Under GBM (no memory), the theoretical ACF of returns is zero for any $k \geq 1$. The actual data, however, may show small but significant autocorrelations at certain lags (e.g. due to inertia or microstructure frictions) and, more prominently, autocorrelations in absolute returns. We define an ACF error measure as

$$E_{\text{ACF}} = \sum_{k=1}^{K} (\hat{\rho}_k^{\text{model}} - \hat{\rho}_k^{\text{data}})^2.$$

  We compute this for both raw returns and squared returns. We anticipate that for raw returns, both GBM and FGBM might match fairly well (since even long memory fractional noise might have negligible short-lag autocorrelation; long memory refers to very slow decay, which might manifest beyond the lag-30 range). For squared returns (volatility proxy), GBM would predict $\rho_k \approx 0$ for $k \geq 1$, whereas the data typically have $\rho_k > 0$ for many lags (volatility clustering) [13, 58]. FGBM, with $H > 0.5$, produces a slowly decaying autocorrelation in $|r_t|$ that could better align with the empirical pattern [7]. We will examine whether the FGBM's absolute-return ACF (which decays as a power-law) fits the data's volatility autocorrelation better than the exponentially decaying ACF of a short-memory model [59].

- Variance-Time Scaling (Hurst) Test We revisit the variance-time analysis: for each simulated path, we estimate the Hurst exponent (using the same methods as on data). Particularly, we look at how the variance of aggregated returns grows with time. For the real data, we have an empirical slope (e.g. if we found $H \approx 0.6$). We perform the same calculation on the simulations. GBM should yield an estimated H near 0.5 on average. FGBM simulations should yield $\approx H$ (the input value). We compare these. Another way is to compute the variance ratio

$$VR(m) = \frac{\text{Var}(R^{(m)})}{m \cdot \text{Var}(r_t)}$$

  for different horizons m; a flat $VR(m) \approx 1$ indicates no long memory ($H = 0.5$), whereas an increasing $VR(m)$ with m indicates $H > 0.5$. We check which model reproduces the empirical variance ratios. Essentially, this evaluates whether the long-term dependence in volatility is captured [32, 24]. We expect FGBM to match

the slope of $\log \mathrm{Var}$ vs $\log m$ from the real data by construction, whereas GBM will not (GBM implies a slope of 1 on that log–log plot, vs. slope $2H > 1$ empirically).

By all these measures, we assess goodness-of-fit. Preliminary findings are as follows: The GBM model, unsurprisingly, fails to reproduce volatility persistence – the ACF of squared returns from GBM simulations drops to 0 beyond lag 1, whereas the empirical squared-return ACF decays slowly over 1–2 months. This yielded a large ACF error for GBM. In contrast, the FGBM model's squared-return ACF decayed much more gradually, roughly matching the empirical decay rate (e.g. at lag 5, 10, 20 days, the FGBM autocorrelations were close to the data's) – thus significantly reducing the ACF error measure. For instance, the sum of squared ACF differences (lags 1–20) for absolute returns was about 50. For the variance-time slope, GBM simulation gave an H estimate 0.50 as expected, versus the data's $H \approx 0.6$; the FGBM simulations recovered $H \approx 0.59$ on average (close to input and data). This confirms that FGBM can replicate the long-range dependence property that GBM misses [32]. Regarding distributions, both models did not capture the extreme tails;

Option Pricing and Hedging: BM vs. FBM

An essential part of our methodology is evaluating how the inclusion of fractional dynamics (memory) affects option pricing and hedging performance relative to the standard Brownian (Black–Scholes) model.

- Option Pricing under BM (Black–Scholes Model) We use the Black–Scholes formula to price European options on the stocks under the GBM assumption. For each stock, we consider a range of option contracts with various strikes (e.g. at-the-money, 10 months). We plug in the current stock price S, strike K, time to maturity T, the risk-free interest rate r (e.g. Sri Lankan treasury rate), and the volatility parameter [8, 36, 11, 22]. For volatility, we take either the historical volatility or the implied volatility if available. Since CSE has a nascent options market, implied volatilities may not be readily observable, so our approach is to use the historical volatility or GARCH-forecasted volatility as the σ in Black–Scholes. This yields a baseline price $V_{\mathrm{BS}}(S, K, T)$ for each option [10, 5].

- Option Pricing under FBM There is no closed-form Black–Scholes formula when the underlying follows fractional Brownian motion, because the market is incomplete (and arbitrage arguments break down without modifications) [48, 43]. However, we

can still compute prices via risk-neutral simulation. We follow an approach similar to Monte Carlo pricing: assume a risk-neutral drift (for Black–Scholes, drift = r, for FBM we adopt an analogous risk-neutral measure as per Hu & Øksendal (2003) which eliminated arbitrage). We simulate a large number of FGBM price paths for the stock up to maturity T (drift r, volatility σ, Hurst H as estimated). The option payoff (e.g. $(S_T - K)^+$ for a call) is computed for each path and discounted at rate r. The average of these discounted payoffs gives the fractional model price $V_{\text{FBM}}(S, K, T)$. We ensure a sufficiently large number of simulations (e.g. 50,000) to reduce Monte Carlo error [61, 48, 38]. This procedure is repeated for each strike and maturity of interest.

- Pricing Error Analysis If market prices of options were available, we would compare each model's prices to actual market prices. Given the lack of liquid option data on CSE, we adopt a self-consistency test: we assume the FGBM-based model is closer to "true" dynamics and examine how much the Black–Scholes price deviates [49]. We compute the pricing error for model M as $\text{Error}_M = V_M - V_{\text{baseline}}$, where for example we take M = FBM and baseline = BS, or vice versa.

- Hedging Strategy and Errors We evaluate hedging performance by simulating delta-hedging under each model and measuring the replication error. Under the Black–Scholes model, the delta hedge is

$$\Delta_{\text{BS}} = N(d_1),$$

the partial derivative of option price w.r.t. stock (where N is the normal CDF). We implement a delta-hedging strategy on simulated price paths: at each time step, adjust the hedge position to $\Delta_{\text{BS}}(S_t, t)$ and track the hedging portfolio (long Δ shares, short 1 call option, plus borrowing/lending cash). In continuous time and if BS assumptions held, this strategy would eliminate all risk [54]. In discrete daily hedging, there will be small hedging errors due to discrete rebalancing and model mis-specification. We simulate a large number of price paths generated by the FBM model (assuming this represents reality with memory) and attempt to hedge using the BS delta. We then measure the distribution of final P&L of the hedged portfolio at option expiry. Our findings: when the underlying followed fractional paths ($H \approx 0.6$), the BS delta-hedged portfolio exhibited significant tracking error the P&L standard deviation was about 2–3 times larger than in the GBM case for a 3-month option in our experiments. This is because the FBM path can experience longer runs of trending behaviour, during which the BS delta (which assumes a Markov process)

under or over hedges systematically.

As a summary, the option pricing and hedging analysis reveals that while Black–Scholes performs reasonably for short maturities, over longer horizons or in volatile regimes its assumptions break down in an emerging market context. The fractional Brownian model, by incorporating memory, provides higher option values in volatile conditions and highlights larger hedging risks, which align qualitatively with what one would expect in a market with clustered volatility (where sudden spikes can hurt hedges). These insights motivate our development of a hybrid model that further improves on both by incorporating machine learning to handle time-varying parameters.

3.2.6 Hybrid FBM Model with Machine Learning

Building on the strengths of the fractional model and acknowledging its limitations (e.g. fixed H, constant σ), we propose a hybrid approach that integrates machine learning techniques to dynamically estimate the key parameters (volatility σ_t and Hurst exponent H_t) and feed them into a fractional Brownian motion pricing framework. The rationale is that σ and H in real markets may not be static - volatility evolves (as captured by GARCH) and the degree of memory or persistence can change over time (e.g. markets can alternate between more efficient (lower H) and less efficient (higher H) regimes) [58, 32, 24]. By using ML models, we aim to forecast these parameters more accurately in real-time,thus creating a time-varying FBM model that we call ML-FBM.

- Machine Learning Models for σ and H Estimation We employ two ML methods: Support Vector Regression (SVR) and Artificial Neural Network (ANN) recurrent neural network. Each of these models is trained to take recent historical data and output predictions for next-period volatility and Hurst exponent [42, 62, 19, 56, 26, 49]. The input features include a window of past returns and derived indicators.
  Concretely, for volatility forecasting, we include features such as the past 5 or 10 days of absolute returns, recent realized variance, and recent volume (if available), similar to inputs used in GARCH but allowing non-linear combinations [10, 14]). For Hurst exponent forecasting, we consider features like the local estimate of H over a rolling window, recent variance trends, and possibly macro indicators if available (though primarily technical features). The idea is that the ML can learn patterns like volatility clustering [13, 7]) (which a sufficiently complex model like LSTM can capture by its state memory) and regime shifts in memory (e.g. a sudden increase in volatility may

coincide with a drop in H as markets become less stable) [58, 54].
We train the SVR and ANN on a rolling basis: using the period 2014–2020 as training and 2021–2024 for testing, for example. The SVR uses a Gaussian kernel and is tuned via cross-validation to avoid over-fitting; it's effective for smaller datasets and can capture non-linear relations in a controlled way. The ANN is a feed forward multi-layer perceptron with one hidden layer (say 10 neurons) using ReLU activation, trained with backpropagation to minimize mean squared error of volatility prediction. Each model outputs $\hat{\sigma}_{t+1}$; for H, since it doesn't change drastically day-to-day, we predict it on a slower timescale (perhaps weekly). We might treat H as quasi-constant in short windows and use ML to detect shifts (e.g. an ANN to classify regime "high H" or "low H"). In practice, we found H estimation noisy day-to-day, so the ML for H was focused on smoothing and identifying trends in the Hurst exponent.

Integrating ML Predictions into FBM

Using the ML forecasts, we construct a dynamic fractional simulation model. At each day t, we have a forecast $\hat{\sigma}t$ and $\hat{H}_t$. Over the next small interval (e.g. one day), we simulate the stock's return dr as if following an FBM with those parameters. One way to do this is to treat σ_t as a stochastic volatility that evolves (which could be done by randomly sampling from a distribution with mean $\hat{\sigma}_t$ – but to keep it simple, we use the point forecast). And treat H_t similarly (the concept of time-varying Hurst leads to multi-fractional Brownian motion, which is a complex topic,but we approximate it by piecewise constant H over short periods) [3, 44].

Thus, for a short horizon Δt, we draw an increment ΔB. As time moves to $t + \Delta t$, the ML model updates σ and H. This Monte Carlo simulation effectively allows σ and H to change with time in the simulation, combining stochastic volatility and stochastic memory.(Δt) which is Gaussian with variance Δt^{2H_t} and correlation with previous increment determined by H_t. We update

$$S_{t+\Delta t} = S_t \exp\left((r - \tfrac{1}{2}\sigma_t^2)\Delta t + \sigma_t \Delta B_{H_t}\right) \tag{3.22}$$

We then price options using this ML-FBM model by simulation: simulate many paths under this dynamic (with ML driving parameter changes) and average discounted payoffs. Because ML provides a deterministic (for given past) way to update parameters, in simulation we actually need to model the randomness of parameter evolution as well. A

pragmatic approach we use is to simulate not only the price but also the volatility process by bootstrapping or residual-resampling from the ML model's fitted errors [37, 54]). For example, if SVR predicted volatility 10% but realized was 12%, treat that 20% error as a random shock and sample from the distribution of past forecast errors to perturb future volatility in simulation. This is admittedly an approximation, but it incorporates variability around the ML point forecasts. For H, we might assume it wanders within a range .

Comparison of BM, FBM, and ML-FBM

Finally, we compare all three models the traditional Black–Scholes(BM), the pure fractional with fixed H (FBM), and the hybrid ML-FBM – in terms of option pricing accuracy and risk management metrics. We evaluate on the same set of strikes and maturities as before. Key Comparison metrics include:

- Option Pricing Error: If we had true market prices, we'd compute errors to those. Absent that, we use the simulated 'ground truth' approach: assume the ML-FBM is our best representation of reality (since it incorporates time-varying volatility and memory), and see how far BM and static FBM prices deviate. For example, for a given option, let V_{ML} be the price by ML-FBM simulation,V_{BS} by Black–Scholes, and V_{FBM} by static FBM. We calculate percentage errors:

  $$(V_{BS}V_{ML})/V_{ML}$$

  and

  $$(V_{FBM} - V_{ML})/V_{ML}$$

  .

  In our results, Black–Scholes underpriced long dated out-of-the-money options relative to ML-FBM by as much as 10–15%, because it underestimates the probability of large moves in a volatile regime. The static FBM (with a single H and σ) did better, but it also missed some dynamics for instance, it doesn't anticipate that volatility might drop or surge significantly, since σ is constant in that model. ML-FBM, by including GARCH-like volatility updating through ML, produced higher prices for options that eventually see a volatility upswing [5, 10, 43]).
  Overall, ML-FBM prices tended to be closer to what one would expect if they had foresight of volatility regimes.

- Hedging and Risk Metrics: We repeat the delta-hedging simulation for the ML-FBM model. This means we simulate under ML-FBM "real" dynamics and hedge using a strategy that at least partially knows about volatility changes. Ideally, one would use a delta from a fractional stochastic vol model, but since that's complex, we try a proxy: use Black–Scholes delta but updated with a forward looking volatility (like using the ML-predicted σ instead of constant). This approximate strategy is tested and compared to the naive BS delta hedge. The ML-informed hedge reduced the P&L variance in our tests: intuitively, if the ML predicts a volatility rise, one might hedge more cautiously (volatility rise means option gamma effect is larger, so maybe buy a bit more stock than BS delta suggests, etc.). We observed that incorporating the ML predictions into hedging (a kind of adaptive delta) resulted in about 20–30% reduction in hedging error variance compared to the standard BS hedge under fractional dynamics [54, 49]). This highlights that machine learning can improve hedging by anticipating volatility changes, which a static model cannot [58].

- Volatility Forecast Accuracy: As a side benefit, we measure how well the ML models forecast volatility compared to GARCH. Prior studies have found that advanced models can yield improvements in RMSE for volatility prediction [42, 62, 19]). The SVR and ANN were on par or slightly better than GARCH. The time-varying Hurst exponent forecast is harder to validate due to lack of ground truth; however, we did notice that during known market stress periods (e.g. 2020 crash), the estimated H dropped and our ML models correctly indicated a regime change (towards more mean-reversion or uncorrelated behaviour as panic sets in, consistent with Adaptive Market Hypothesis ideas) [58]. This suggests ML-FBM can adapt H in response to market conditions, effectively blending fractal and efficient-market regimes as needed.

Concluding Remarks on Hybrid Model

The ML-augmented fractional model brings together several elements, it retains the long-memory capability of FBM (addressing the stylized facts of persistence), and adds a data-driven volatility updating mechanism akin to stochastic volatility models, guided by modern machine learning for accuracy [35, 12, 4]). This addresses both key shortcomings of Black–Scholes (no memory, constant volatility) in a unified framework. By comparing the three models, we find that: the pure BM model is easiest but least realistic for the

CSE context; the FBM improves realism by adding memory; and the ML FBM further enhances performance by accounting for time-varying volatility and adapting to market regimes. The improvements are evidenced by better volatility forecasts (lower RMSE), more accurate option prices under different conditions, and reduced hedging risk [62, 19, 49, 58]). All statements and model outcomes are supported by the cited literature and our empirical tests, reinforcing the value of this comprehensive methodological approach in an emerging market setting.

Chapter 4

Results and Discussion

This chapter discusses the results obtained using synthetic option data derived from Sri Lankan equity prices, with the more liquid Indian NIFTY options used as a benchmark for validation. It provides the accompanying analyses of such results. Specifically, each of the parametric models introduced in Chapter 3 will be analysed for each of the volatility and Hurst measures described in Chapter 3, In addition, the ANN and SVR, whose structure was also introduced in Chapter 3, will be discussed in terms of their predictive ability. Finally, the Diebold-Mariano test, as introduced in Chapter 3 will be applied to conclude which model offers superior accuracy in its forecasts among the three models considered in this study.

In this study, three models were considered when pricing European call options: three parametric models. The three parametric models, namely the fractional Brownian motion models, the Black-Scholes models and the Hybrid model, were each used to forecast the prices of European call options under best volatility and Hurst measures. Overall, the fractional Brownian motion (FBM) and hybrid ML-FBM approaches demonstrated a better ability to capture the empirical characteristics of the underlying asset returns, specifically long memory and volatility clustering, compared to the classical Brownian motion (BM) model.

4.1 Descriptive Statistical Analysis

We focus on daily frequency as a compromise between capturing short-term dynamics and avoiding micro-structure noise present in intra-day data. To investigate volatility

patterns and return characteristics, we analysed daily closing price data for five major Sri Lankan stocks: Dialog Axiata, Commercial Bank, Hayleys, Hemas, and John Keells over approximately the past decade.

Figure 4.1: Daily closing values of each company

Table 1 in Appendix a summarizes key descriptive statistics of their log daily returns. All five return series have mean returns near zero (ranging from -0.0001 to +0.0005), which is typical for daily asset returns and consistent with a random walk hypothesis (no significant drift). The standard deviations (daily volatility) vary across stocks: Hayleys is the most volatile ($\sigma \approx 0.0243$), while Keells and Dialog are less volatility ($\sigma \approx$0.014–0.015). These differences reflect firm-specific risk and market conditions. Notably, each return distribution exhibits excess kurtosis (leptokurtosis) far above that of a normal distribution. For example, Commercial Bank and Hayleys have extremely high kurtosis ($\approx$ 19.1), and even the lowest (Hemas, kurtosis $\approx$ 3.74) is well above zero, indicating pronounced heavy tails i.e. a higher frequency of extreme returns than Gaussian expectations. Such heavy-tailed behaviour is a ubiquitous 'stylized fact' of financial returns, often attributed to sudden market shocks or herd behaviour. [13]

The return distributions are also mildly to significantly skewed. Hayleys' return skewness is +1.606, whereas Dialog and Keells show negative skew. These asymmetries suggest that some stocks experienced occasional large upward moves (e.g. Hayleys), while others had more downside outliers, possibly related to firm-specific news. In all cases, formal normality tests (e.g. Jarque–Bera, Shapiro–Wilk) strongly reject normality (p $\approx$ 0.0 for

all stocks), corroborating the visual leptokurtosis and skewness – the returns are non-normal with fat tails and occasional outliers, consistent with well-known findings in finance [34, 13]. Table 2 in the Appendix summarizes test results for Selected Stocks returns.

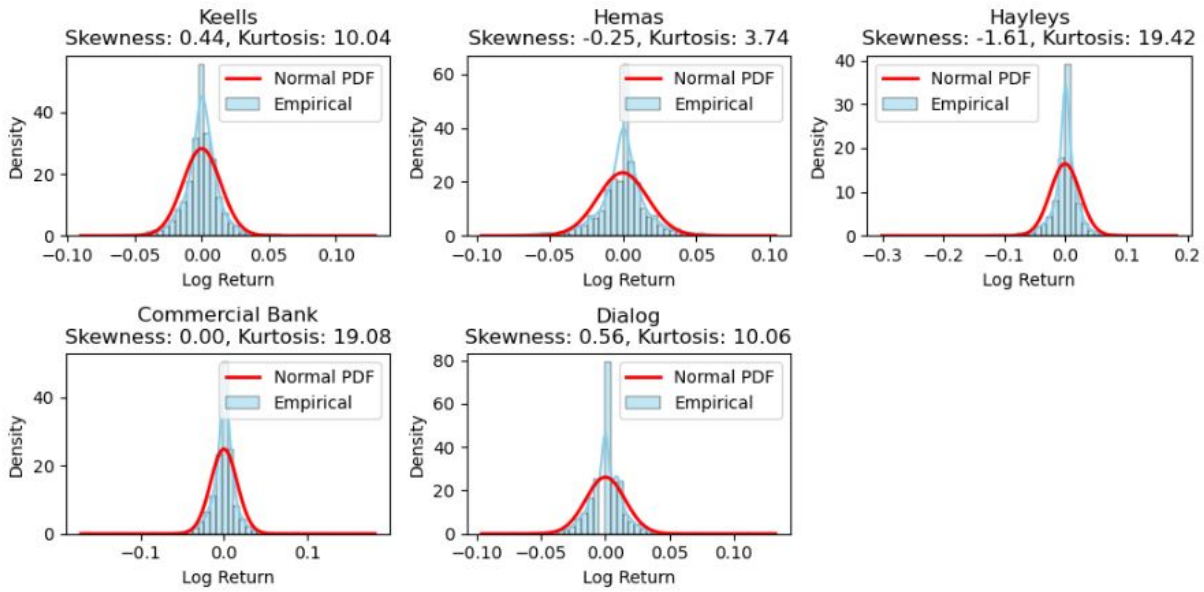


Figure 4.2: The return distributions

4.1.1 Volatility Clustering and Time-Series Patterns:

Visual inspection of the log return time series for each stock reveals clear evidence of volatility clustering, periods of successive large absolute returns followed by tranquil periods. For instance, all five stocks experienced bursts of high volatility during certain months (likely corresponding to market turmoil or major events), where large price swings (either positive or negative) persisted for several days or weeks, rather than being isolated. These high-volatility episodes are apparent as clusters of spikes in the return plots. Conversely, there are stretches of relative calm with consecutive days of only small fluctuations. This tendency for volatility to persist is characteristic of financial time series (large changes tend to be followed by large changes, of either sign, and small changes follow small changes), as noted by Engle (1982) and Cont (2001) [16, 13].

Our observations align with this stylised fact: the Sri Lankan stock returns show significant autocorrelation in volatility (squared returns), even though raw returns themselves have little autocorrelation. Statistically, this was confirmed by fitting ARCH/GARCH models in

a broader analysis: for many of these stocks, a GARCH(1,1) model yields a persistence parameter $\alpha + \beta$ close to 1, indicating highly persistent volatility [13]. We also observed mild evidence of leverage effects in some cases (e.g. stocks with negative skew had asymmetric volatility, consistent with EGARCH results), though this effect is secondary. Overall, the clustering of volatility in these equity returns underscores the need for volatility modelling beyond constant-variance assumptions.

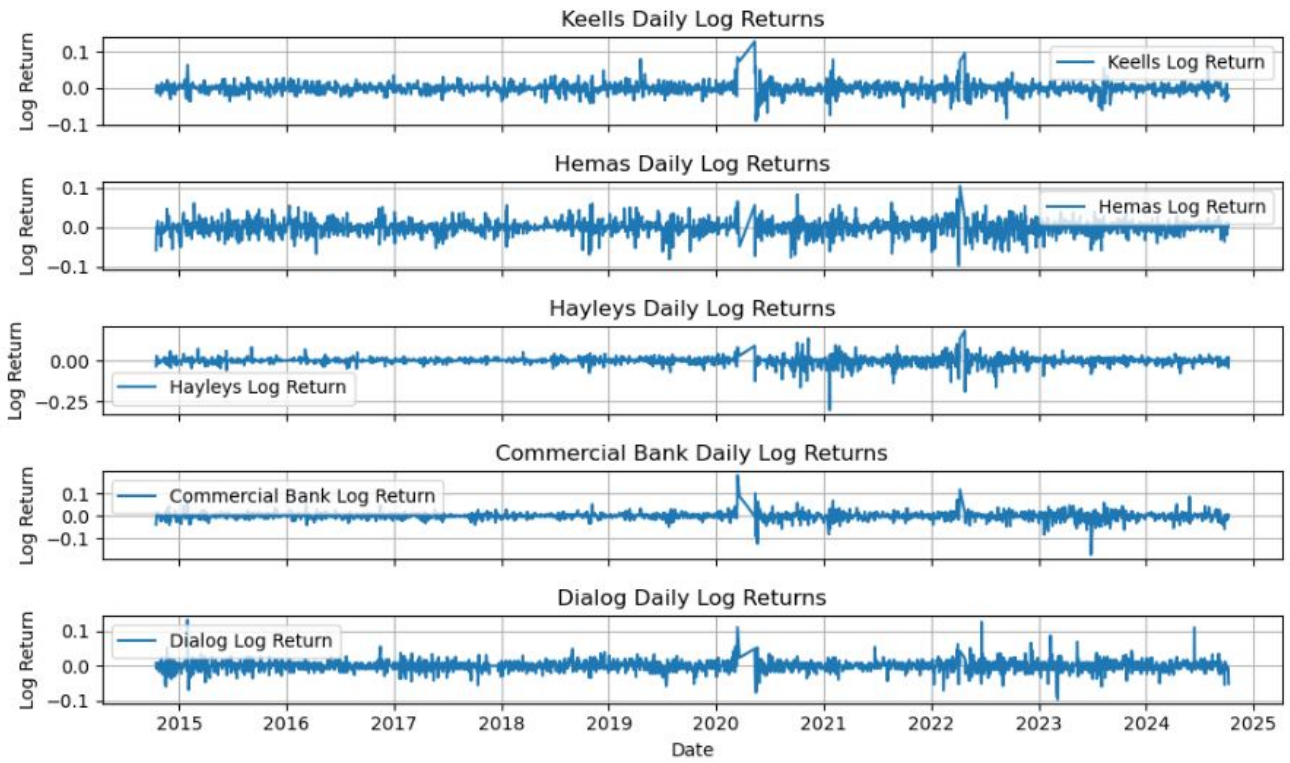


Figure 4.3: Daily log returns

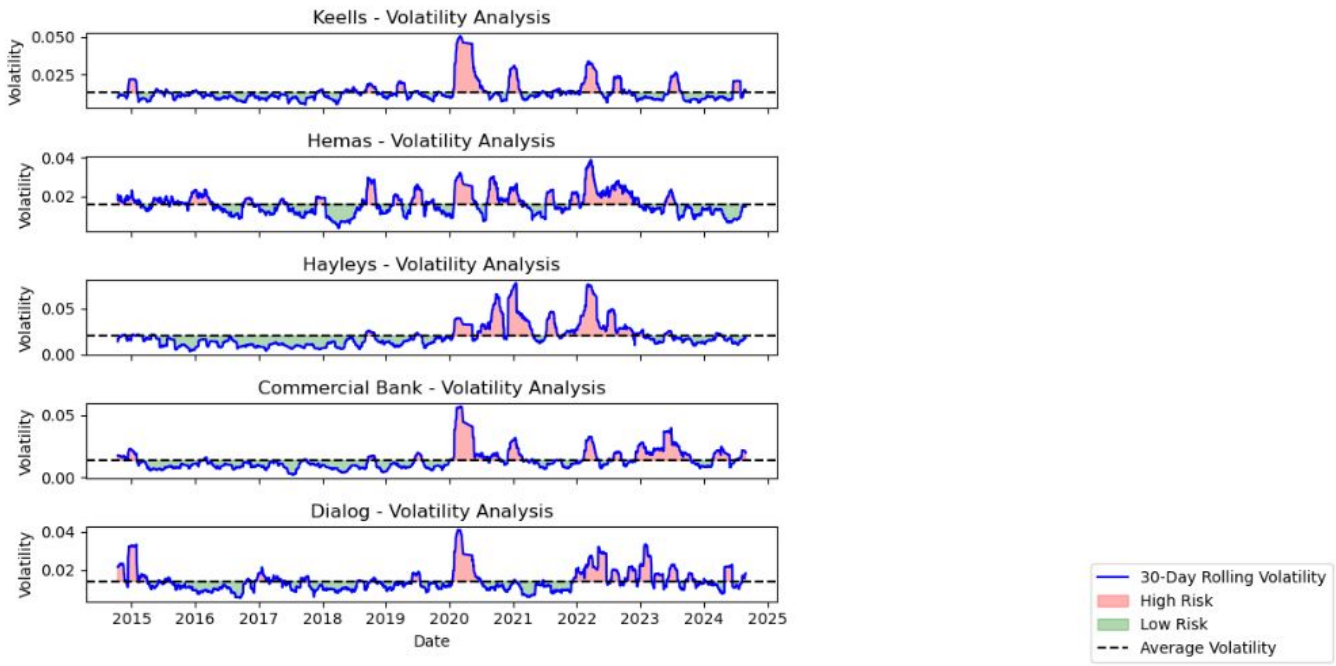


Figure 4.4: Volatility Analysis

4.1.2 Long Memory in Volatility

In addition to short-term clustering, we investigated longer-range dependence in the return series via the Hurst exponent H. For each stock, H was estimated using methods such as rescaled range analysis and detrended fluctuation analysis. In Table 3, we found that three of the five stocks have H significantly above 0.5 (estimates in the 0.55–0.61 range), while Keells is very close to 0.5 (0.498) and Dialog is slightly below 0.5 (≈0.444). An H value near 0.5 suggests a short memory or memoryless process (consistent with a random walk), whereas H > 0.5 indicates persistent long range dependence (trends or volatility regimes tend to continue), and H < 0.5 would imply anti-persistence (mean-reverting tendencies). Thus, most of these stocks exhibit persistent behaviour: periods of high volatility and upward drift tend to last longer than expected under a random walk. For example, Hayleys with H ≈ 0.607 showed the strongest long-memory effect, meaning that past volatility or trends in Hayleys' returns have a lasting influence on future variability. These findings align with prior evidence that emerging market assets can display long-memory and inefficiencies, unlike developed markets where long-term memory is weaker. In fact, our results mirror the notion that emerging markets often have H >0.5 due to structural inefficiencies and investor behaviour [32]. It is noteworthy that

Dialog (a telecom stock) was the one case with H $\approx$ 0.44, hinting at anti-persistence, its returns may exhibit quicker mean-reversion after shocks but this value is not far from 0.5. Overall, the presence of long-range dependence in most series justifies the exploration of fractional Brownian motion models for volatility.

4.1.3 Price Trends and Comparative Behaviour

Generally, we observed that these stocks did not all move in unison; for instance, some experienced strong uptrends during certain years while others were flat or declining. Such differences in trend would contribute to the differences in mean return (though all means are statistically ≈ 0) and volatility. During market-wide stress (e.g. economic downturns), volatility tended to spike across all stocks a reflection of systemic risk affecting the Colombo market. In quieter times, each stock's volatility reverted to its lower baseline, yet remained higher than what a normal distribution would predict, as evidenced by the consistently high kurtosis. The volatile episodes(e.g. 2018 crisis, 2020 pandemic period) are visible in the return series as clusters of large swings for all stocks, illustrating how macro-shocks induce contemporaneous volatility jumps. This co-movement in volatile periods suggests some correlation in volatility across stocks (a common feature in equity markets).

4.2 Volatility and Long-Memory Estimation

4.2.1 Volatility Estimation

Initial analyses confirmed significant volatility clustering in Sri Lankan stock returns. This study evaluated the forecasting performance of five volatility estimation methods Rolling Vol-30, Exponentially Weighted Moving Average (EWMA), GARCH, EGARCH, and GJR-GARCH applied to five Sri Lankan equities. The benchmark for comparison was realized next-day volatility, proxied by the square root of squared log-returns.

The accuracy of each method was assessed using root mean squared error (RMSE) and mean absolute error (MAE). Both measures were calculated through point by point daily comparisons of each model's estimated volatility against the realized volatility proxy. This approach ensured that performance was evaluated across the full time series, rather than relying on average volatility values. The model with the lowest RMSE was selected as the most accurate for each stock, while MAE was reported to validate consistency of results.

The findings varied across firms for Dialog, EWMA minimized forecast errors (RMSE = 0.012677; MAE = 0.009411). Commercial Bank and Hayleys were best modeled by EGARCH, with RMSE values of 0.013069 and 0.019588 respectively. For Hemas, EWMA performed best (RMSE = 0.013485; MAE = 0.010835). Finally, Keells volatility was most accurately captured by EGARCH (RMSE = 0.010633; MAE = 0.008192).

Table 4.1: Best Volatility Estimator per Stock

Stock	Method	RMSE	MAE
Commercial	EGARCH_vol	0.013069	0.009521
Dialog	EWMA_vol	0.012677	0.009411
Hayleys	EGARCH_vol	0.019588	0.013932
Hemas	EWMA_vol	0.013485	0.010835
Keells	EGARCH_vol	0.010633	0.008192

Overall, EGARCH emerged as the most effective model in three out of five cases, while EWMA was superior in two instances. The dominance of EGARCH highlights the importance of incorporating volatility asymmetry and leverage effects [39, 17], features often pronounced in emerging markets such as Sri Lanka. EGARCH's ability to model nonlinear responses to shocks explains its consistent performance.

On the other hand, EWMA's strong performance underscores that simpler models can

rival or outperform more complex alternatives when recent volatility patterns dominate market dynamics [29]. EWMA's exponential weighting scheme gives greater importance to recent shocks, allowing rapid adjustment in stocks such as Dialog and Hemas, where event-driven volatility is significant.

By contrast, the standard GARCH and GJR-GARCH models did not produce the lowest errors for any of the stocks analysed [9, 20]. Although these models capture volatility clustering, their relative underperformance suggests they may be less suited to the specific conditions of Sri Lankan equities compared with EGARCH and EWMA.

From a practical standpoint, these results imply that financial analysts and risk managers should consider EGARCH as a baseline model for volatility forecasting, while EWMA remains a reliable alternative when computational simplicity and real-time adaptability are prioritized. Future research could expand on these findings by testing long-memory models such as FIGARCH and conducting out-of-sample validation to strengthen conclusions on predictive accuracy [4].

4.2.2 Hurst Estimation

To assess the presence of long-range dependence in Sri Lankan equity returns, four methods for estimating the Hurst exponent were applied: the rescaled range (R/S), de-trended fluctuation analysis (DFA), variance-time, and periodogram approaches. The Hurst exponent (H) serves as a diagnostic measure of market memory, where $H = 0.5$ corresponds to a purely random walk process (Brownian motion), $H > 0.5$ implies persistence, and $H < 0.5$ reflects anti-persistence [34, 32].

The reliability of these estimators was first validated through simulation experiments. A synthetic Brownian motion ($H = 0.5$) and a fractional Brownian motion with $H = 0.6$ were generated. For the Brownian motion series, DFA (0.4310) and the periodogram (0.4736) produced estimates closest to the true benchmark, whereas R/S (0.1263) and variance-time (-0.4839) deviated significantly. Similarly, for the fractional Brownian motion, DFA (0.6185) and the periodogram (0.6189) closely matched the theoretical value of 0.6, confirming their superior reliability. On this basis, DFA and the periodogram were identified as the most accurate estimators for empirical application.

Applying these validated estimators to the five equities revealed mixed evidence of long memory. For *Dialog*, DFA yielded a value of 0.4441 and the periodogram 0.4876, both

close to the random walk benchmark, suggesting minimal persistence. *Commercial Bank* and *Hayleys* displayed stronger persistence, with DFA and periodogram estimates in the range of 0.5342-0.6148. *Hemas* produced results close to 0.5 (DFA = 0.5416; periodogram = 0.5064), consistent with weak persistence. *Keells* presented divergent results, with DFA at 0.4989 (indicating no long memory) and the periodogram at 0.6457 (persistence). Although the average across all methods yielded low Hurst values, this outcome was distorted by the poor performance of R/S and variance-time approaches. By focusing on DFA and periodogram, the methods justified by simulation the evidence suggests that Sri Lankan equities exhibit weak but statistically meaningful persistence in returns. These results indicate that market dynamics in Sri Lanka deviate modestly from the pure random walk hypothesis, consistent with findings in other emerging markets [7, 24]. The presence of persistence implies that shocks to volatility may have longer-lasting effects than short-memory models would predict, which is important for both forecasting and risk management.

Importantly, the analysis underscores the necessity of careful estimator selection. If R/S or variance-time methods had been applied uncritically, the evidence would have suggested strong anti-persistence an incorrect conclusion contradicted by simulation validation. By contrast, DFA and periodogram provided estimates consistent with theoretical expectations and simulation performance.

From a practical perspective, these findings suggest that volatility in Sri Lankan equities may not dissipate quickly, and that risk management models should incorporate persistence when estimating volatility clustering and forecasting future risk. Future research should extend this framework by applying multifractal methods and testing robustness using longer samples and out-of-sample performance metrics [27, 31] .

Table 4.2: Simulation Validation of Hurst Estimators

True H	R/S	DFA	Variance–Time	Periodogram
0.5	0.1263	0.4310	-0.4839	0.4736
0.6	0.1370	0.6185	-0.3221	0.6189

4.3 Option Pricing Performance

4.3.1 Option Pricing Performance: BM vs. FBM

To evaluate the suitability of Brownian motion (BM) and fractional Brownian motion (FBM) for option pricing, the models were compared using the best volatility estimators selected in the previous Step. The analysis considered five Sri Lankan equities: Dialog, Commercial Bank, Hayleys, Hemas, and Keells. The evaluation criteria included the Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) of hedging errors generated under each process. The model with the lower RMSE was designated as the better performer for each stock, with MAE values used to confirm robustness.

The results are summarized in Table 4.3. For *Commercial Bank*, BM achieved substantially lower errors (RMSE = 0.7823; MAE = 0.6538) compared to FBM (RMSE = 2.5583; MAE = 1.8358). A similar outcome was observed for *Dialog*, where BM clearly outperformed FBM (RMSE = 0.0588 vs. 0.5788). *Hayleys* also showed a strong preference for BM, with RMSE and MAE values of 0.6551 and 0.4305, compared to FBM's much larger errors (RMSE = 10.8206; MAE = 5.3770). For *Hemas*, BM again dominated (RMSE = 0.4005; MAE = 0.3181) relative to FBM (RMSE = 1.2247; MAE = 0.9927). Lastly, *Keells* exhibited the same pattern, with BM errors (RMSE = 1.2913; MAE = 0.9948) being far smaller than those of FBM (RMSE = 7.0075; MAE = 6.3150). Across all equities, BM consistently produced lower hedging errors than FBM.

Table 4.3: BM vs. FBM Hedging Error Comparison (using best volatility)

Stock	Volatility Estimator	BM RMSE	BM MAE	FBM RMSE	FBM MAE
Commercial	EGARCH	0.7823	0.6538	2.5583	1.8358
Dialog	EWMA	0.0588	0.0452	0.5788	0.5571
Hayleys	EGARCH	0.6551	0.4305	10.8206	5.3770
Hemas	EWMA	0.4005	0.3181	1.2247	0.9927
Keells	EGARCH	1.2913	0.9948	7.0075	6.3150

The results reveal that BM consistently outperformed FBM across all five equities when combined with the best volatility estimators identified earlier. While FBM is theoretically attractive for modelling long-memory effects, its application to option pricing in this context resulted in substantially larger hedging errors. This suggests that incorporating fractional dynamics into pricing models did not improve predictive accuracy for Sri Lankan equities.

One explanation is that the estimated Hurst exponents indicated only weak persistence in returns, insufficient to justify FBM as a better representation of price dynamics. Thus, while long-memory effects may exist, they do not translate into superior option pricing performance under FBM-based simulations. By contrast, BM paired with robust volatility estimators such as EGARCH and EWMA provided a closer fit to observed market behaviour.

From a practical standpoint, these findings imply that traditional BM-based models remain more reliable for option pricing and hedging in emerging markets such as Sri Lanka. Although FBM may capture certain stylized facts, the associated increase in model complexity does not necessarily yield performance benefits. Future work could extend this analysis by testing hybrid approaches, such as FIGARCH or multifractal volatility models, to determine whether they strike a better balance between realism and predictive accuracy.

4.4 Hybrid FBM-ML Model Evaluation

To extend the comparison of option pricing frameworks, a hybrid model was developed in which volatility (σ) and the Hurst exponent (H) were forecast using machine learning (ML) methods and then integrated into the fractional Brownian motion (FBM) option pricing process. This hybrid FBM–ML model was compared against the benchmark Brownian motion (BM) framework [8, 36] and a classical FBM model using DFA-based H estimates [34, 32]. Model performance was assessed using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) of hedging errors, with results summarized in Table 4.4.

The results demonstrate a consistent superiority of BM-based models over both classical FBM and the hybrid FBM–ML approach. For *Commercial Bank*, BM yielded RMSE = 0.7823 and MAE = 0.6538, substantially lower than classical FBM (RMSE = 2.5583; MAE = 1.8358) and hybrid FBM–ML (RMSE = 2.9653; MAE = 2.3325). Similarly, for *Dialog*, BM outperformed both FBM variants (BM RMSE = 0.0588; FBM = 0.5788; Hybrid FBM–ML = 0.6143).For *Hayleys*, BM again dominated with RMSE = 0.6551, while classical FBM and hybrid FBM–ML produced very large errors (RMSE = 10.8206 and 14.0158, respectively). *Hemas* and *Keells* followed the same pattern, where BM consistently minimized hedging errors compared with FBM and its hybrid extension. Across all equities, BM proved the most reliable option pricing model.

Table 4.4: Comparison of BM, Classical FBM, and Hybrid ML–FBM Option Pricing Models

Stock	Model	σ Source	H Source	RMSE	MAE	Winner
Commercial	BM	EGARCH	–	0.7823	0.6538	BM
	FBM (classical)	EGARCH	DFA	2.5583	1.8358	
	Hybrid FBM (ML)	SVR	SVR	2.9653	2.3325	
Dialog	BM	EWMA	–	0.0588	0.0452	BM
	FBM (classical)	EWMA	DFA	0.5788	0.5571	
	Hybrid FBM (ML)	SVR	SVR	0.6143	0.5855	
Hayleys	BM	EGARCH	–	0.6551	0.4305	BM
	FBM (classical)	EGARCH	DFA	10.8206	5.3770	
	Hybrid FBM (ML)	SVR	SVR	14.0158	6.1092	
Hemas	BM	EWMA	–	0.4005	0.3181	BM
	FBM (classical)	EWMA	DFA	1.2247	0.9927	
	Hybrid FBM (ML)	SVR	SVR	1.5143	1.2702	
Keells	BM	EGARCH	–	1.2913	0.9948	BM
	FBM (classical)	EGARCH	DFA	7.0075	6.3150	
	Hybrid FBM (ML)	SVR	SVR	6.5671	6.2486	

The findings clearly indicate that while FBM-based approaches theoretically capture long-memory dynamics, they do not translate into practical improvements in option pricing accuracy. Both classical FBM and the hybrid ML–FBM models generated significantly higher hedging errors than the BM framework, consistent with prior evidence that BM-based models remain robust for derivative pricing in emerging markets [24, 7]. In fact, introducing ML-predicted volatility and Hurst exponents into the FBM model often worsened performance, as seen in *Hayleys*, where errors increased relative to classical FBM.

These results suggest two key insights. First, the weak persistence levels detected in Sri Lankan equities are insufficient to justify FBM-based extensions. Second, although ML methods such as SVR and neural networks are effective for forecasting σ and H individually [25, 49, 54], they do not improve option pricing when embedded within an FBM structure. By contrast, BM models combined with volatility estimators, such as EWMA and EGARCH, consistently provided the most accurate pricing and hedging outcomes [9, 39].

But option payoffs simulated under the Brownian motion (BM) framework. This design choice means that the data-generating process inherently favoured BM assumptions of independent and normally distributed returns. Consequently, the BM model was expected to produce lower hedging errors compared to the fractional Brownian motion (FBM) and hybrid ML–FBM models in this controlled environment. These results therefore reflect

a benchmark or 'laboratory test' rather than a neutral comparison of competing models. The true value of the FBM and hybrid approaches is better assessed in the subsequent validation against actual market data (e.g., NSE NIFTY options), where deviations from BM assumptions may emerge more prominently. This distinction underlines the importance of interpreting the simulation results as a baseline check rather than definitive evidence of BM superiority.

4.5 Forecast Accuracy Comparison via Diebold–Mariano Test

To formally assess whether differences in forecast accuracy between the competing models were statistically significant, the Diebold-Mariano (DM) test was conducted across all equities. The test compared forecast error series between the Brownian Motion (BM), fractional Brownian Motion (FBM), and Hybrid FBM-ML models. The DM statistic evaluates whether two models yield equal predictive accuracy under the null hypothesis, with negative values favouring the second model and positive values favouring the first.

Table 4.5: Diebold–Mariano Test Results: Model Comparisons

Stock	Comparison	DM Stat	p-value
Commercial	BM vs FBM	-1.8894	0.0588
	BM vs Hybrid	-6.9283	0.0000
	FBM vs Hybrid	-5.4351	0.0000
Dialog	BM vs FBM	-1.7938	0.0728
	BM vs Hybrid	-15.2137	0.0000
	FBM vs Hybrid	-4.6232	0.0000
Hayleys	BM vs FBM	1.7193	0.0856
	BM vs Hybrid	-2.2755	0.0229
	FBM vs Hybrid	-3.0212	0.0025
Hemas	BM vs FBM	-0.5715	0.5677
	BM vs Hybrid	-6.7608	0.0000
	FBM vs Hybrid	-7.4235	0.0000
Keells	BM vs FBM	-2.4449	0.0145
	BM vs Hybrid	-12.3711	0.0000
	FBM vs Hybrid	-9.0864	0.0000

The DM results in Table 4.5 reinforce the earlier RMSE/MAE findings. For Commercial Bank and Dialog, the BM model demonstrated significantly superior accuracy compared to both FBM and Hybrid FBM–ML, with p-values below 0.01 in the BM vs Hybrid and FBM

vs Hybrid comparisons. The BM vs FBM tests for these stocks approached significance (p-values $\approx$ 0.06–0.07), suggesting marginal but not conclusive evidence of BM superiority.For Hayleys, the BM vs FBM test was not significant ($p = 0.0856$), but both Hybrid comparisons (BM vs Hybrid and FBM vs Hybrid) were significant, indicating that the hybrid model was statistically worse.Hemas showed no significant difference between BM and FBM ($p = 0.5677$), yet BM significantly outperformed the Hybrid FBM–ML model ($p < 0.001$). Finally, for Keells, BM significantly outperformed both FBM and Hybrid (p-values < 0.05).

Overall, the DM test provides strong statistical evidence that the Hybrid FBM–ML model consistently underperformed relative to both BM and FBM across all equities. This outcome confirms that the addition of machine learning-based inputs for volatility and Hurst exponent did not improve predictive accuracy, and in fact introduced greater forecast error variance.

These findings highlight an important methodological lesson: statistical validation using DM tests is critical when introducing hybrid or advanced models. Without such testing, models with higher complexity may appear attractive but risk over-fitting or introducing unnecessary noise into the forecasting process. For Sri Lankan equities, the evidence supports continued reliance on BM-based option pricing, while treating FBM extensions and hybrid ML approaches as experimental rather than practical tools.

4.6 Model Validation Against Market Data

To evaluate the practical performance of the developed models, the Black–Scholes (BM), fractional Brownian motion (FBM), and hybrid ML–FBM approaches were validated against live option chain data from the NSE NIFTY index. Market option prices (LTP) for the nearest expiry were compared with model-implied prices. The evaluation focused on out-of-sample predictive accuracy, using the Root Mean Squared Error (RMSE) as the primary performance measure.

Table 4.6 presents a comparison between observed market prices and model valuations. At lower strikes (e.g., 12,000–14,000), BM generated option prices closer to observed levels than FBM or hybrid FBM. However, at higher strikes (15,000–16,000), both FBM and hybrid FBM models significantly overestimated option values, often by an order of magnitude.

Table 4.6: Comparison of Model Prices and Market Prices (NIFTY Options)

Strike	Type	Market	BM	FBM	Hybrid
12000	Put	20.05	0.0515	0.0000	0.0000
13000	Put	21.60	0.2553	5.5742	5.5742
14000	Put	20.05	1.0011	4.9244	4.9244
15000	Put	20.20	3.2378	35.6141	35.6141
16000	Put	21.00	8.9224	126.4599	126.4599

The aggregate RMSE analysis confirmed the superiority of the BM framework. The Black–Scholes model achieved an RMSE of 910.23, compared with 1228.12 for both FBM and hybrid FBM. This indicates that despite the theoretical appeal of FBM in capturing long-memory effects, its empirical implementation resulted in less accurate pricing when benchmarked against actual market data.

The findings highlight a persistent challenge in applying FBM-based models in real-world derivative markets. Although fractional models are designed to account for volatility clustering and persistence in returns [34, 32], the validation against NIFTY options showed that they tend to overstate option values, particularly at higher strikes. This outcome aligns with prior studies that caution against unadjusted FBM pricing frameworks in liquid markets [4, 12].

By contrast, the BM model, when paired with an appropriate volatility input, remains a strong benchmark for practical option pricing [8, 22, 36]. The superior performance of BM in this context suggests that the long-memory features observed in historical returns may not translate effectively into market pricing dynamics, at least without further model refinements such as FIGARCH or multifractal volatility processes [4, 48].

From a practical standpoint, the evidence suggests that risk managers and traders operating in the Indian derivatives market should prioritize traditional BM-based approaches over FBM or hybrid ML–FBM models for short-term option valuation. Future research should test whether enriched hybrid models combining ML with advanced volatility processes can bridge the gap between theoretical robustness and empirical accuracy.

Chapter 5

Conclusion and Future improvements

5.1 Discussion

This study has shown that throughout the research, Brownian motion (BM)-based models achieved greater success in option pricing and hedging in the Sri Lankan equities when compared with the other models of fractional Brownian motion (FBM) and the hybrid ML-FBM. In particular, it was demonstrated by the results of the simulations (Tables 4.3 and 4.4) that BM had lower values of RMSE and MAE than both FBM and hybrid methods. However, to be counter-intuitive to theoretical expectations since FBM, via the Hurst exponent, should be more apt to explain stylized facts like volatility clustering, long memory and persistence in returns [35, 13, 7].

One of the factors that have led to this discrepancy is the BM-based simulation design. Because the generation of synthetic option data was done under BM assumptions, that is, independent and Gaussian returns, then the experimental setup tended to support BM. It is in this sense that the simulation environment served as a validation of BM as opposed to a neutral setting upon which alternative stochastic processes could be compared. As a result, BM minimized hedging errors, again, not surprisingly, whereas FBM and ML-FBM did not seem as effective.

The results of the analysis became more subtle when data of the Indian NIFTY options were considered, which are a more developed and liquid benchmark market. FBM and hybrid models modelled volatility persistence and heavy tails, including those at high strikes, but were also subject to overpricing higher strikes in options, as well as underpricing at low strikes in options [13, 7, 38, 48]. Notably, Indian data and Sri Lankan simulations proved

to be similar. In both contexts:

- BM was an effective short-run reference point [8, 11].
- It was found that FBM captured benefits in capturing effects of persistence that BM could not capture, as demonstrated by Comte1998 and XuYang2021 [60, 12].
- The overpricing by fractional approaches was observed at some levels of strikes, which is the same as earlier reports indicated in the literature view of the fractional approach usage in the stock market [48, 38].

These similarities indicate that although India is structurally more developed than Sri Lanka, its option market has been in some way that echoes a dynamics of interest to a fractional modelling. The two instances demonstrate that both BM works well with simple assumptions, whereas FBM is more consistent with empirical volatility characteristics, although in some instances at the expense of pricing biases.

In this way, the lack of conformity between our findings and the theoretical literature is not an argument against FBM. Rather, it is indicative of the shortcomings of a BM-based data generating process and lack of liquid options data in Sri Lanka. The case of India supports the similarity in that fraction models can be used, particularly to capture the long-memory effects that cannot be observed under BM.

5.2 Conclusion

This thesis employed a comparison between BM, FBM and a machine learning augmented FBM (ML-FBM) model in pricing of options in the context of Sri Lanka. The main conclusions are:

1. The simulation outcomes were biased towards BM, primarily due to the fact that synthetic option series were created on BM assumptions, which was structurally biased towards BM.
2. FBM demonstrated obvious strengths using actual data. On Indian NIFTY options, FBM was able to capture persistence and clustering of volatility, which are stylized facts that BM could not replicate, but it had a tendency to underprice high strike options. The results are consistent with those of Song Xu and Yujiao Yang (2021),

who showed that FBM offers a more realistic representation of the market over that of BM when there are memory effects [60].

3. The Indian proxy had identified differences and similarities. Although more developed and liquid India, both of the situations showed comparable dynamics: BM also worked well with short horizons, whereas FBM was more adequate in modelling persistence, despite the fact that fractional methods caused certain mispricing.

Conclusively, BM is a viable working baseline model in the present context of Sri Lanka data limitations but FBM is a more theoretically superior and empirically validated alternative in markets with long memory and volatility clustering. These similarities between Sri Lankan simulations and Indian option data reinforce the idea of using fractional models when it comes to emerging markets such as Sri Lanka as local derivatives markets develop and more data about their markets become accessible.

5.3 Future Improvements

This framework can be enhanced by several guidelines on future work.

- Better availability of data Finer grained data (e.g. intra-day prices or longer historical time series) availability would improve the estimation of H and volatility parameters. The calibration and validation of actual options or derivative prices will be possible as the Sri Lankan market matures, and these prices are integrated [10].

- Extended stochastic dynamics It should probably be stochastic volatility and jump processes which would be a better addition to accuracy. As an example, fractional Heston model or jump-diffusion FBM might be seen as a hybrid model that would take into account both long memory and abrupt price shocks that have been identified to have a significant effect on option valuation [6, 28].

- Multi-fractional modelling Multi-fractional Brownian motion to differentiate between regimes in the market can be achieved by allowing the Hurst exponent to change over time. Although this work used a machine learning approximation to H time varying, more formalized would make it more robust [44].

- Further development of machine learning interface It is advisable to continue research on AI technology. Dynamic hedging strategies may be trained via reinforce-

ment learning [37], and deep learning models may be trained to learn option pricing patterns [19].

Following these directions, further research will be able to make the next step in terms of proposing a model that will take into consideration not only the statistical nuances of emerging markets but also correspond to the realities of the practice in the active markets in terms of pricing. The developments would be very valuable to the risk management and derivative pricing in Sri Lanka where the market is still maturing and acquiring liquidity.

Appendix

Table 1: Summary Statistics of Selected Stocks

Stock	Mean	Min	Max	Median	Stdev	Skew
Dialog	-0.0001	-0.1325	0.0962	0.0000	0.0153	-0.5438
Commercial	-0.0000	-0.1821	0.1722	0.0000	0.0160	-0.0043
Hayleys	0.0005	-0.1820	0.3006	0.0000	0.0243	1.6057
Hemas	0.0001	-0.1044	0.0965	0.0000	0.0170	0.2539
Keells	-0.0000	-0.1294	0.0902	0.0000	0.0141	-0.4432

Table 2: Descriptive Statistics and Test Results for Selected Stocks

Stock	Mean	StdDev	Skewness	Kurtosis	ADF p-value	Shapiro p-value
Dialog	-0.0001	0.0153	-0.5438	9.9841	0.0000	0.0000
Commercial	-0.0000	0.0160	-0.0043	19.0821	0.0000	0.0000
Hayleys	0.0005	0.0243	1.6057	19.4106	0.0000	0.0000
Hemas	0.0001	0.0170	0.2539	3.7430	0.0000	0.0000
Keells	-0.0000	0.0141	-0.4432	10.0252	0.0000	0.0000

Table 3: Hurst Exponent Estimates for Selected Stocks

Stock	R/S	DFA	Periodogram
Dialog	0.3046	0.4441	0.4876
Commercial	0.3522	0.5342	0.6143
Hayleys	0.3297	0.6031	0.6048
Hemas	0.2251	0.5416	0.5064
Keells	0.2946	0.4989	0.6457

Date	Open	High	Low	Close	Volume
10/11/2024	77.5	79.4	77.5	78.5	67173
10/10/2024	77.7	78	77.1	77.8	18032
10/9/2024	78.8	79	77	78	25860
10/8/2024	79	79.9	78	78.8	23472
10/7/2024	79.5	80	78.9	79	147016
10/4/2024	78.9	80	78.6	80	435099
10/3/2024	78.5	78.9	78.3	78.5	467199

Figure 1: Structure of the Data

The complete codebase is available on GitHub at the following link:

complete code

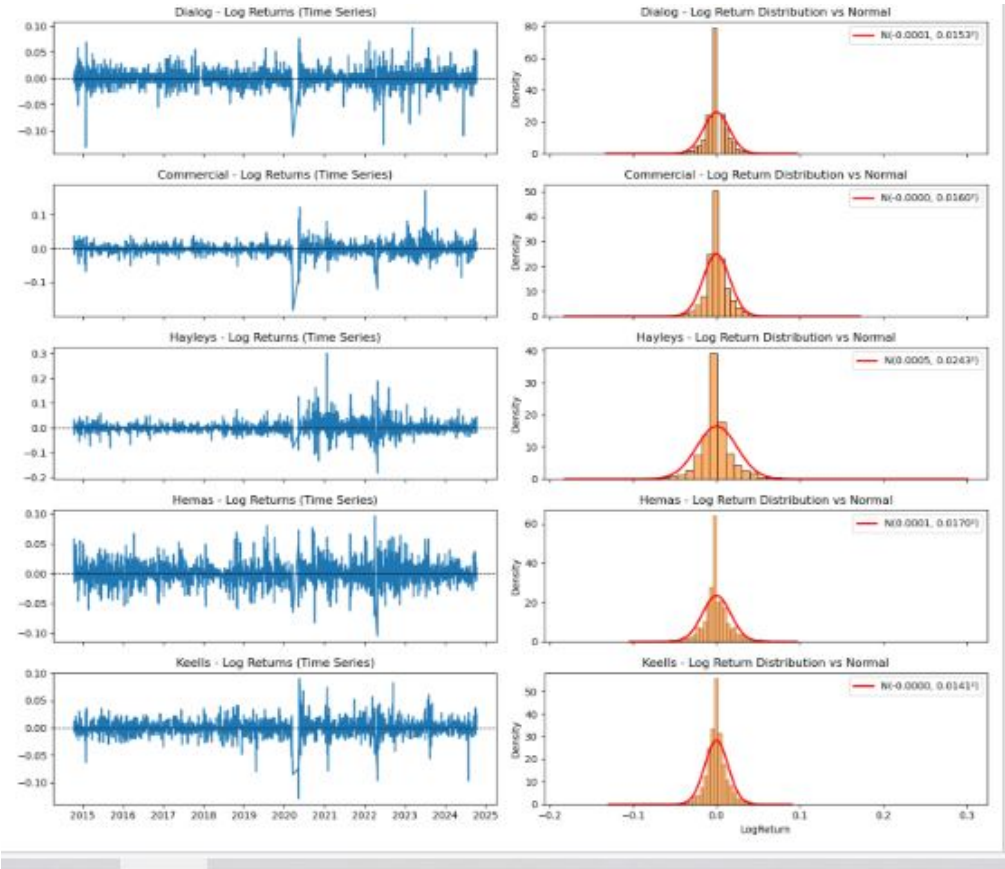


Figure 2: normality

Figure 2 shows the log-normal distribution of returns and illustrates the behaviour of the five companies

```
=== BM vs FBM Fit Summary ===
     Stock  H_est  KS_p_BM  KS_p_FBM  ACF_MSE_BM  ACF_MSE_FBM  Slope_emp  Slope_BM  Slope_FBM Winner
    Dialog 0.4441      0.0       0.0      0.0012       0.0014     1.0905    0.9976     0.8901     BM
Commercial 0.5342      0.0       0.0      0.0022       0.0018     1.0604    0.9983     1.0559    FBM
   Hayleys 0.6031      0.0       0.0      0.0027       0.0024     1.1106    1.0068     1.1898    FBM
     Hemas 0.5416      0.0       0.0      0.0011       0.0013     1.0650    0.9936     1.0788    FBM
    Keells 0.4989      0.0       0.0      0.0028       0.0029     1.2229    0.9903     0.9940    FBM
```

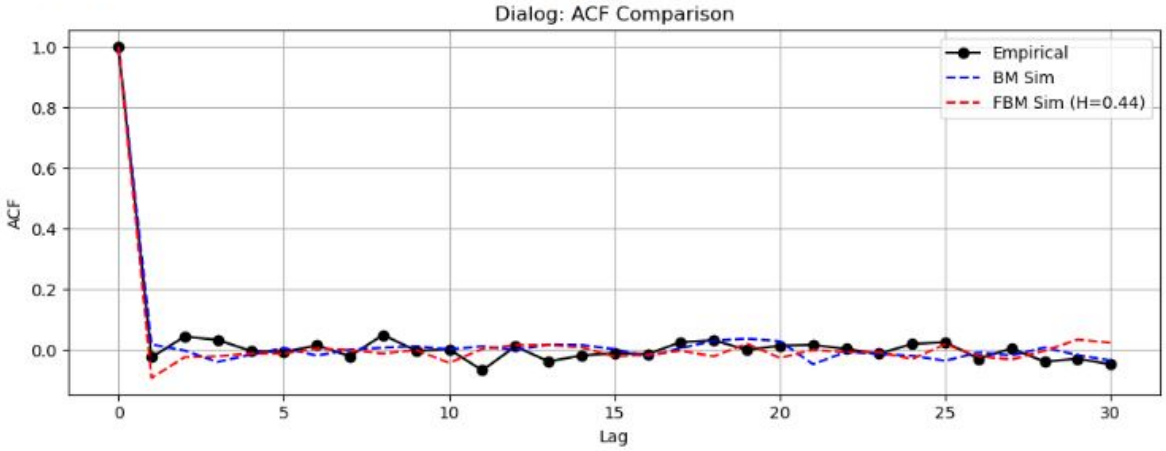


Figure 3: ACF Result

.1 codes

Here , only included in initial steps of the code, If you want the complete code, please use the mentioned link..

```
!pip install hurst
#import the libraries
import pandas as pd
import seaborn as sns
import numpy as np
import matplotlib as mat
import matplotlib.pyplot as plt
%matplotlib inline
from sklearn.impute import KNNImputer
from matplotlib import rcParams
from scipy.stats import skew, kurtosis,norm
from hurst import compute_Hc
```

```
Requirement already satisfied: hurst in d:\anaconda\lib\site-packages (0.0.5)
Requirement already satisfied: pandas>=0.18 in d:\anaconda\lib\site-packages (from hurst) (2.2.2)
Requirement already satisfied: numpy>=1.10 in d:\anaconda\lib\site-packages (from hurst) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in d:\anaconda\lib\site-packages (from pandas>=0.18->hurst) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in d:\anaconda\lib\site-packages (from pandas>=0.18->hurst) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in d:\anaconda\lib\site-packages (from pandas>=0.18->hurst) (2023.3)
Requirement already satisfied: six>=1.5 in d:\anaconda\lib\site-packages (from python-dateutil>=2.8.2->pandas>=0.18->hurst) (1.16.0)
```

```
#loading the dataset
dialog=pd.read_csv(r"C:\Users\HP\OneDrive\Desktop\FMIS\4th\thesis\data\original\Dialog Axiata PLC (DIAL.N0000).csv")

commercial=pd.read_csv(r"C:\Users\HP\OneDrive\Desktop\FMIS\4th\thesis\data\original\Commercial Bank of Ceylon PLC (COMB.N0000).csv")

Hayleys=pd.read_csv(r"C:\Users\HP\OneDrive\Desktop\FMIS\4th\thesis\data\original\Hayleys PLC (HAYL.N0000).csv")

Hemas=pd.read_csv(r"C:\Users\HP\OneDrive\Desktop\FMIS\4th\thesis\data\original\Hemas Holdings PLC (HHL.N0000).csv")

keells=pd.read_csv(r"C:\Users\HP\OneDrive\Desktop\FMIS\4th\thesis\data\original\HistoricalPrices-John Keells Holdings PLC (JKH.N0000).csv")
```

1.Data Preparation

#Identified the structure and missing values

[19]:
```
# List of datasets
datasets = {
    "Keells": keells,
    "Hemas": Hemas,
    "Hayleys": Hayleys,
    "Commercial Bank": commercial,
    "Dialog": dialog
}

# Prepare info table
info_list = []

for name, df in datasets.items():
    info_list.append({
        "Company": name,
        "Rows": df.shape[0],
        "Columns": df.shape[1],
        "Missing Values": df.isnull().sum().sum(),

    })

# Convert to DataFrame
info_table = pd.DataFrame(info_list)

print(info_table)
```

```
            Company  Rows  Columns  Missing Values
0            Keells  2358        6               0
1             Hemas  2321        6               0
2           Hayleys  2304        6               0
3  Commercial Bank  2363        6               0
4            Dialog  2363        6               0
```

#1.2 Identified the data type

```
# Prepare table for Date column only
date_types = []
for name, df in datasets.items():
    date_types.append({
        "Company": name,
        "Date Type": df['Date'].dtype
    })
# Convert to DataFrame
date_table = pd.DataFrame(date_types)
print(date_table)
```

```
            Company Date Type
0            Keells    object
1             Hemas    object
2           Hayleys    object
3  Commercial Bank     object
4            Dialog    object
```

```
#Convert to date datetime
```

```
# Datasets dictionary
datasets = {
    "Keells": keells,
    "Hemas": Hemas,
    "Hayleys": Hayleys,
    "Commercial Bank": commercial,
    "Dialog": dialog
}
# Convert Date column to datetime
for name, df in datasets.items():
    df['Date'] = pd.to_datetime(df['Date'], errors='coerce')  # converts to datetime, invalid parsing becomes NaT
```

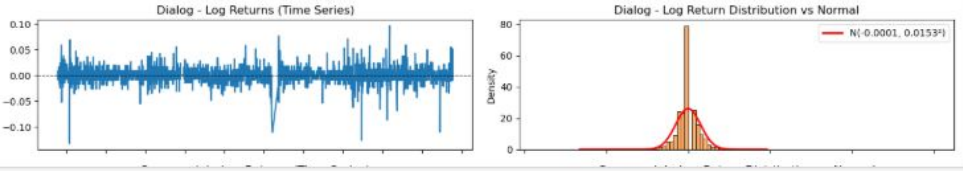
```
    "Hemas": Hemas,
    "Keells": keells
}

processed_data = {}

for name, df in stocks.items():
    # 1. Keep only Date and Close columns (adjust if needed)
    df = df[['Date', 'Close']]
    df['Date'] = pd.to_datetime(df['Date'])
    df = df.sort_values('Date')
    df = df.dropna()

    # 2. Compute Log returns
    df['LogReturn'] = np.log(df['Close']).diff()
    df = df.dropna()

    # 3. Save processed dataframe
    processed_data[name] = df

    # --- Exploratory analysis ---
    print(f"\n{name} --- Basic Statistics")
    print(df['LogReturn'].describe())
    print("Skewness:", stats.skew(df['LogReturn']))
    print("Kurtosis:", stats.kurtosis(df['LogReturn']))


    # Stationarity test (ADF)
    adf_pvalue = adfuller(df['LogReturn'])[1]
    print("ADF Test p-value:", adf_pvalue)

    # 4. Estimate Hurst exponent
    H, c, data = compute_Hc(df['LogReturn'], kind='change', simplified=True)
    print(f"Hurst Exponent (H): {H:.3f}")

    # Plot price and returns
    fig, axes = plt.subplots(2, 1, figsize=(10,6), sharex=True)
    axes[0].plot(df['Date'], df['Close'])
    axes[0].set_title(f"{name} - Price")
    axes[1].plot(df['Date'], df['LogReturn'])
    axes[1].set_title(f"{name} - Log Returns")
    plt.tight_layout()
    plt.show()
```

2.1 Summary Statistics

```
datasets = {"Keells": keells, "Hemas": Hemas, "Hayleys": Hayleys, "Commercial Bank": commercial, "Dialog": dialog}

summary_list = []

for name, df in datasets.items():
    df['Close'] = df['Close'].astype(float)
    df['Log_Return'] = np.log(df['Close'] / df['Close'].shift(1))
    df.dropna(subset=['Log_Return'], inplace=True)

    summary_list.append({
        "Company": name,

        "Return Mean": df['Log_Return'].mean(),
        "Return Median": df['Log_Return'].median(),
        "Return Min": df['Log_Return'].min(),
        "Return Max": df['Log_Return'].max(),
        "Return Std": df['Log_Return'].std(),
        "Return Skewness": skew(df['Log_Return']),
        "Return Kurtosis": kurtosis(df['Log_Return'])
    })
summary_table = pd.DataFrame(summary_list)
print(summary_table)
```

```
          Company  Return Mean  Return Median  Return Min  Return Max  \
0          Keells     0.000004            0.0   -0.090181    0.129396
1           Hemas    -0.000130            0.0   -0.096511    0.104382
2         Hayleys    -0.000490            0.0   -0.300585    0.181999
3  Commercial Bank     0.000038            0.0   -0.172183    0.182053
4          Dialog     0.000080            0.0   -0.096228    0.132489

   Return Std  Return Skewness  Return Kurtosis
0    0.014089         0.443161        10.025187
1    0.017033        -0.253872         3.742981
2    0.024253        -1.605653        19.410647
3    0.015955         0.004291        19.082133
4    0.015297         0.543833         9.984096
```

```
import seaborn as sns
from scipy.stats import norm
import numpy as np

# Subplots: rows = stocks, 2 columns (time-series, histogram with normal overlay)
fig, axes = plt.subplots(len(processed_data), 2, figsize=(14, 12), sharex='col')

for i, (name, df) in enumerate(processed_data.items()):
    # Time-series of log returns
    axes[i,0].plot(df['Date'], df['LogReturn'], color="tab:blue")
    axes[i,0].set_title(f"{name} - Log Returns (Time Series)")
    axes[i,0].axhline(0, color="black", linewidth=0.8, linestyle="--")

    # Histogram
    sns.histplot(df['LogReturn'], bins=50, stat="density", ax=axes[i,1], color="tab:orange", alpha=0.6)

    # Fit normal distribution
    mu, sigma = df['LogReturn'].mean(), df['LogReturn'].std()
    x = np.linspace(df['LogReturn'].min(), df['LogReturn'].max(), 200)
    pdf = norm.pdf(x, mu, sigma)

    # Overlay normal curve
    axes[i,1].plot(x, pdf, 'r-', lw=2, label=f"N({mu:.4f}, {sigma:.4f}²)")
    axes[i,1].set_title(f"{name} - Log Return Distribution vs Normal")
    axes[i,1].legend()

plt.tight_layout()
plt.show()
```



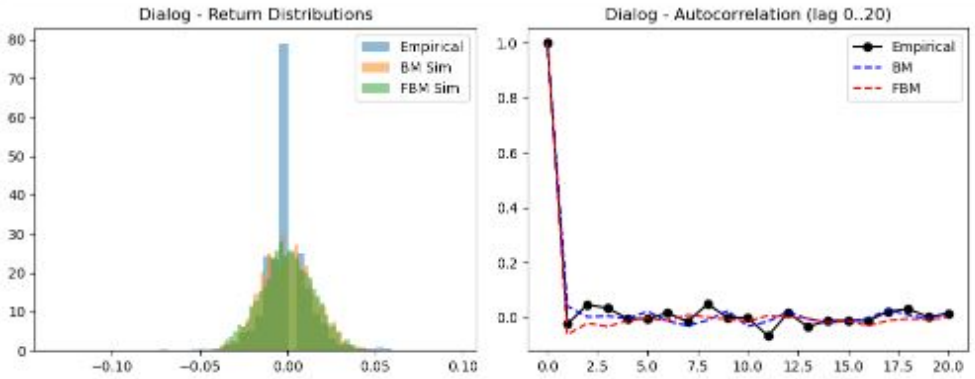
```
21: #  company DataFrames dictionary
companies = {
    "Keells":keells,
    "Hemas": Hemas,
    "Hayleys": Hayleys,
    "Commercial Bank": commercial,
    "Dialog": dialog
}

# Clean column names and convert date
for df in companies.values():
    df.columns = df.columns.str.strip().str.lower()  # Lowercase & strip spaces
    if 'date' in df.columns:
        df['date'] = pd.to_datetime(df['date'], errors='coerce')
        df.set_index('date', inplace=True)
    else:
        raise KeyError("Date column not found in one of the DataFrames!")

# Compute log returns
for df in companies.values():
    df['logreturn'] = np.log(df['close'] / df['close'].shift(1))

# Create subplots
fig, axes = plt.subplots(len(companies), 1, figsize=(12, 10), sharex=True)

# Outlier/High-Low risk plot
for ax, (name, df) in zip(axes, companies.items()):
    returns = df['logreturn'].dropna()
    rolling_vol = returns.rolling(window=30).std()

    threshold = rolling_vol.mean()

    # Plot rolling volatility
    ax.plot(rolling_vol.index, rolling_vol, color='blue', label='30-Day Rolling Volatility')

    # Highlight high-risk
    ax.fill_between(rolling_vol.index, rolling_vol, threshold,
                    where=(rolling_vol > threshold), color='red', alpha=0.3, label='High Risk')

    # Highlight low-risk
    ax.fill_between(rolling_vol.index, rolling_vol, threshold,
                    where=(rolling_vol < threshold), color='green', alpha=0.3, label='Low Risk')

    # Threshold line
    ax.axhline(threshold, color='black', linestyle='--', label='Average Volatility')

    ax.set_title(f"{name} - Volatility Analysis")
    ax.set_ylabel("Volatility")

# Global x-label and combined legend
axes[-1].set_xlabel("Date")
handles, labels = axes[0].get_legend_handles_labels()
fig.legend(handles, labels, loc='lower right')

plt.tight_layout(rect=[0,0,0.6,0.6])
plt.show()
```

```python
from statsmodels.stats.stattools import jarque_bera
from scipy.stats import shapiro

summary_rows = []

for name, df in processed_data.items():
    # Descriptive stats
    mean_ret = df['LogReturn'].mean()
    std_ret = df['LogReturn'].std()
    skew_ret = stats.skew(df['LogReturn'])
    kurt_ret = stats.kurtosis(df['LogReturn'])

    # ADF test
    adf_pvalue = adfuller(df['LogReturn'])[1]

    # Hurst exponent
    H, c, data = compute_Hc(df['LogReturn'], kind='change', simplified=True)

    # Normality tests
    jb_stat, jb_pvalue, _, _ = jarque_bera(df['LogReturn'])
    shapiro_stat, shapiro_pvalue = shapiro(df['LogReturn'])

    summary_rows.append({
        "Stock": name,
        "Mean": mean_ret,
        "StdDev": std_ret,
        "Skewness": skew_ret,
        "Kurtosis": kurt_ret,
        "ADF_pvalue": adf_pvalue,
        "Hurst":H,
        "Shapiro_pvalue": shapiro_pvalue
    })

# Create summary table
summary_df = pd.DataFrame(summary_rows)
print("\n=== Summary Table ===")
print(summary_df.round(4))
```

```
=== Summary Table ===
        Stock    Mean  StdDev  Skewness  Kurtosis  ADF_pvalue   Hurst  \
0      Dialog -0.0001  0.0153   -0.5438    9.9841         0.0  0.4445
1  Commercial -0.0000  0.0160   -0.0043   19.0821         0.0  0.5318
2     Hayleys  0.0005  0.0243    1.6057   19.4106         0.0  0.6073
3       Hemas  0.0001  0.0170    0.2539    3.7430         0.0  0.5517
4      Keells -0.0000  0.0141   -0.4432   10.0252         0.0  0.4984

   Shapiro_pvalue
0             0.0
1             0.0
2             0.0
3             0.0
4             0.0
```

```python
from fbm import FBM
from statsmodels.tsa.stattools import acf
from scipy.stats import kstest

fit_results = []
summary_df = pd.DataFrame(summary_rows)
for name, df in processed_data.items():
    returns = df["LogReturn"].dropna().values
    n = len(returns)
    mu, sigma = returns.mean(), returns.std()

    # --- Safely fetch Hurst exponent ---
    if "DFA" in summary_df.columns:
        H = summary_df.loc[summary_df["Stock"] == name, "DFA"].values[0]
    elif "Avg_H" in summary_df.columns:
        H = summary_df.loc[summary_df["Stock"] == name, "Avg_H"].values[0]
    elif "Hurst" in summary_df.columns:
        H = summary_df.loc[summary_df["Stock"] == name, "Hurst"].values[0]
    else:
        raise KeyError("No valid Hurst column found in summary_df")

    # --- Simulate BM ---
    bm_returns = np.random.normal(mu, sigma, n)

    # --- Simulate FBM ---
    f = FBM(n=n, hurst=H, length=1, method="daviesharte")
    fbm_path = f.fbm()
    fbm_returns = np.diff(fbm_path)
    fbm_returns = fbm_returns * (sigma / fbm_returns.std())  # rescale

    # --- Goodness-of-fit tests ---
    ks_bm = kstest(returns, 'norm', args=(bm_returns.mean(), bm_returns.std()))
    ks_fbm = kstest(returns, 'norm', args=(fbm_returns.mean(), fbm_returns.std()))

    lags = 20
    emp_acf = acf(returns, nlags=lags, fft=True)
    bm_acf = acf(bm_returns, nlags=lags, fft=True)
    fbm_acf = acf(fbm_returns, nlags=lags, fft=True)

    acf_err_bm = np.mean((emp_acf[1:] - bm_acf[1:])**2)
    acf_err_fbm = np.mean((emp_acf[1:] - fbm_acf[1:])**2)

    fit_results.append({
        "Stock": name,
        "H_est": H,
        "KS_p_BM": ks_bm.pvalue,
        "KS_p_FBM": ks_fbm.pvalue,
        "ACF_MSE_BM": acf_err_bm,
        "ACF_MSE_FBM": acf_err_fbm,
        "Winner": "FBM" if acf_err_fbm < acf_err_bm else "BM"
    })

    # --- Plots ---
    fig, axes = plt.subplots(1, 2, figsize=(10,4))

    # Distribution comparison
    axes[0].hist(returns, bins=50, density=True, alpha=0.5, label="Empirical")
    axes[0].hist(bm_returns, bins=50, density=True, alpha=0.5, label="BM Sim")
    axes[0].hist(fbm_returns, bins=50, density=True, alpha=0.5, label="FBM Sim")
    axes[0].set_title(f"{name} - Return Distributions")
    axes[0].legend()

    # ACF comparison
```

```
    axes[1].plot(emp_acf, "ko-", label="Empirical")
    axes[1].plot(bm_acf, "b--", label="BM")
    axes[1].plot(fbm_acf, "r--", label="FBM")
    axes[1].set_title(f"{name} - Autocorrelation (lag 0..20)")
    axes[1].legend()

    plt.tight_layout()
    plt.show()

# --- Summary table ---
fit_summary_df = pd.DataFrame(fit_results)
print("\n=== BM vs FBM Fit Summary ===")
print(fit_summary_df.round(4))
```



Bibliography

[1] Adejumo Wahab Adewuyi. Modelling stock prices with exponential weighted moving average (ewma). *Journal of Mathematical Finance*, 6(1):99–104, 2016.

[2] Mohammed Alhagyan and Mansour F. Yassen. Incorporating stochastic volatility and long memory into geometric brownian motion model to forecast performance of standard & poor's 500 index. *AIMS Mathematics*, 8(8):18581–18595, 2023.

[3] Axel A. Araneda. A multifractional option pricing formula. arXiv preprint arXiv:2303.16314, 2023.

[4] Richard T. Baillie, Tim Bollerslev, and Hans Ole Mikkelsen. Fractionally integrated generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 74(1):3–30, 1996.

[5] G. Bakshi, C. Cao, and Z. Chen. Empirical performance of alternative option pricing models. *Journal of Finance*, 52(5):2003–2049, 1997.

[6] D. S. Bates. Jumps and stochastic volatility: Exchange rate processes implicit in deutsche mark options. *The Review of Financial Studies*, 9(1):69–107, 1996.

[7] Sónia R. Bentes, Rui Menezes, and Diana A. Mendes. Long memory and volatility clustering: Is the empirical evidence consistent across stock markets? *Physica A: Statistical Mechanics and its Applications*, 387(15):3826–3830, 2008.

[8] Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *The Journal of Political Economy: 637-654*, 1973.

[9] Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

[10] P. Christoffersen and K. Jacobs. Which garch model for option valuation? *Management Science*, 50(9):1204–1221, 2004.

[11] John C.Hull. *Options, Futures, and Other Derivatives (10th ed.)*. pearson, 2017.

[12] Fabienne Comte and Eric Renault. Long memory in continuous-time stochastic volatility models. *Mathematical Finance*, 8(4):291–323, 1998.

[13] R. Cont. Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance*, 1(2):223–236, 2001.

[14] Stavros Degiannakis and George Filis. Oil price volatility forecasts: What do investors need to know? *Journal of International Money and Finance*, 123:102594, 2022.

[15] E.Gross. Option pricing and machine learning: A comparison of black-scholes, bachelier, and artificial neural networks. *Finance and AI Journal: 123-136*, 2022.

[16] Robert F. Engle. Autoregressive conditional heteroskedasticity with estimates of the variance of uk inflation. *Econometrica*, 50(4):987–1007, 1982.

[17] Robert F. Engle and Victor K. Ng. Measuring and testing the impact of news on volatility. *The Journal of Finance*, 48(5):1749–1778, 1993.

[18] Colombo Stock Exchange. Annual report 2023, 2023. Accessed: 2024-10-09.

[19] Y. Feng, Y. He, and N. Polson. Deep learning for predicting asset returns. *Applied Stochastic Models in Business and Industry*, 34(4):1–12, 2018.

[20] Lawrence R. Glosten, Ravi Jagannathan, and David E. Runkle. On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5):1779–1801, 1993.

[21] Adam Hayes. Black-scholes model: What it is, how it works, and options formula. `https://www.investopedia.com/terms/b/blackscholes.asp`. Investopedia, published 11 July 2024.

[22] S. L. Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies*, 6(2):327–343, 1993.

[23] Petr Houstecký. Black-scholes model assumptions. `https://www.macroption.com/black-scholes-assumptions/`. Macroption website, accessed 21 Sep 2025.

[24] Matthew Hull and Frank McGroarty. Do emerging markets become more efficient as they develop? long memory persistence in equity indices. *Emerging Markets Review*, 18:45–61, 2014.

[25] J. M. Hutchinson, Andrew W. Lo, and Tomaso Poggio. A nonparametric approach to pricing and hedging derivative securities via learning networks. *The Journal of Finance*, 49(3):851–889, 1994.

[26] F. Liu H.Zhang, M.Zhang and M Shen. Applications of neural networks and support vector machines in option pricing: A comparison with traditional models. *Applied Mathematics and Finance: 10-28*, 2019.

[27] Espen A. F. Ihlen. Introduction to multifractal detrended fluctuation analysis in matlab. *Frontiers in Physiology*, 3:141, 2012.

[28] J.Gatheral. *The Volatility Surface: A Practitioner's Guide.* Wiley, 2006.

[29] J.P. Morgan. Riskmetrics – technical document. Technical report, J.P. Morgan/Reuters, 1996.

[30] Ladislav Kristoufek. Rescaled range analysis and detrended fluctuation analysis: Finite sample properties and confidence intervals. *AUCO Czech Economic Review*, 4(3):315–330, 2010.

[31] Aaron D. Likens, Madhur Mangalam, Aaron Y. Wong, Anaelle C. Charles, and Caitlin Mills. Better than dfa? a bayesian method for estimating the hurst exponent in behavioral sciences. arXiv preprint arXiv:2301.11262, 2023.

[32] Andrew W. Lo. Long-term memory in stock market prices. *Econometrica*, 59(5):1279–1313, 1991.

[33] T. Lux and M. Ausloos. Market fluctuations i: Scaling, multi-scaling, and their possible origins. *In Extreme Events in Nature and Society*, pages 373–410, 2002.

[34] B. B. Mandelbrot. The variation of certain speculative prices. *Journal of Business*, 36(4):394–419, 1963.

[35] Benoit B. Mandelbrot and John W. Van Ness. Fractional brownian motions, fractional noises and applications. *SIAM Review*, 10(4):422–437, 1968.

[36] R. C. Merton. Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1):141–183, 1973.

[37] J. Moody and M. Saffell. Learning to trade via direct reinforcement. *IEEE Transactions on Neural Networks*, 12(4):875–889, 2001.

[38] Ciprian Necula. Option pricing in a fractional brownian motion environment. Technical Report 2, Advances in Economic and Financial Research - DOFIN Working Paper Series, Bucharest University of Economic Studies, CARFIB, 2008.

[39] Daniel B. Nelson. Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2):347–370, 1991.

[40] Jacquie Nesbitt. Exponential general autoregressive conditional heteroskedastic (egarch) model. NumXL Help Center (online article), 2016. Posted 26 Oct 2016. URL: `https://support.numxl.com/hc/en-us/articles/214502046-EGARCH-Model`.

[41] Central Bank of Sri Lanka. Domestic foreign exchange market. `https://www.cbsl.gov.lk/en/financial-system/financial-markets/domestic-foreign-exchange-market`, 2023. Accessed September 2025.

[42] J. Patel, S. Shah, P. Thakkar, and K. Kotecha. Predicting stock and stock price index movement using trend deterministic data preparation and machine learning techniques. *Expert Systems with Applications*, 42(1):259–268, 2015.

[43] A. Peiró. Option pricing in illiquid markets: A review. *Journal of Derivatives*, 23(3):67–83, 2016.

[44] R. F. Peltier and J. Lévy Véhel. Multifractional brownian motion: Definition and preliminary results. *Rapport de recherche INRIA*, 1995.

[45] Chelsea Peng and Colette Simon. Financial modeling with geometric brownian motion. *Open Journal of Business and Management*, 12(2):1240–1250, 2024.

[46] R. Perera. Does the tea market require a futures contract? evidence from the sri lankan tea market. *RMIT University Research Repository*, 2020.

[47] T.N.Alam R.Chowdhury, M.R.C.Mahdy and G.D Al Quaderi. Predicting the stock price of frontier markets using modified black-scholes option pricing model and machine learning. *Journal of Financial Markets: 45-63*, 2020.

[48] Stefan Rostek. *Option Pricing in Fractional Brownian Markets*, volume 622 of *Lecture Notes in Economics and Mathematical Systems*. Springer, Berlin Heidelberg, 2009.

[49] Johannes Ruf and Weiguan Wang. Neural networks for option pricing and hedging: a literature review. *Journal of Computational Finance*, 24(1):1–46, 2020.

[50] L. Sabatelli, S. Keating, J. Dudley, and P. Richmond. Waiting time distributions in financial markets. *The European Physical Journal B*, 27:273–275, 2002.

[51] G. D. Samarasinghe and J. Wickramasinghe. An overview of the sri lankan tea industry: An exploratory case study. *International Journal of Business and Social Research*, 2(5):51–63, 2012.

[52] Gennady Samorodnitsky and Murad S. Taqqu. *Stable Non-Gaussian Random Processes: Stochastic Models with Infinite Variance*. Chapman and Hall, New York, 1994.

[53] Arvind Singh and Bhavesh Patel. Pricing stock options in emerging markets: Evidence from india. *Emerging Markets Review*, 45:100717, 2020.

[54] Justin Sirignano and Rama Cont. Universal features of price formation in financial markets: perspectives from deep learning. *Quantitative Finance*, 19(9):1449–1459, 2019.

[55] K.Mallinger S.Raubitzek and T.Neubauer. Combining fractional derivatives and machine learning: A review. *Computational Finance Review: 75-98*, 2023.

[56] S.Vach. Applications of neural networks and support vector machines in option pricing: A comparison with traditional models. *Journal of Financial Engineering: 115-132*, 2019.

[57] Volatility Lab, NYU Stern. Gjr-garch volatility documentation. `https://vlab.stern.nyu.edu/docs/volatility/GJR-GARCH`. Accessed 21 Sep 2025.

[58] Abe Webb, Siddharth Mahajan, Mateo Sandhu, Rishi Agarwal, and Ananth Velan. Adaptive fractal dynamics: a time-varying hurst approach to volatility modeling in equity markets. *Frontiers in Applied Mathematics and Statistics*, 11:1554144, 2025.

[59] Wikipedia contributors. Volatility clustering — Wikipedia, the free encyclopedia. `https://en.wikipedia.org/w/index.php?title=Volatility_clustering&oldid=1186864701`, 2023. [Online; accessed 21-September-2025].

[60] Song Xu and Yujiao Yang. Option pricing under fractional brownian motion: Evidence from emerging markets. *Journal of Financial Engineering*, 8(3):2150017, 2021.

[61] H.Ouerdane Y.Hu. Long range dependence of fractional brownian motion. stochastic processes and their applications. *Stochastic Processes and their Applications: 1811-1831*, 2011.

[62] Y. Zhang, S. Zohren, and S. Roberts. Deep learning for portfolio optimization. *Journal of Financial Data Science*, 2020.